Physics 7501 (Autumn 2026)
Quantum Mechanics 1 (graduate core course)

Yuan-Ming Lu

September 4, 2026

Abstract. In this elementary course of quantum mechanics, we will learn the mathematical basis of quantum mechanics (i.e. linear algebra), the postulates of quantum theory, and how to use quantum mechanics to understand microscopic quantum phenomena.

Contents

0 Course Information

Instructor

Professor **Yuan-Ming Lu**

Contact Information

- **Email:** lu.1435@osu.edu
- **Office Phone:** 614-292-3084
- **Office:** 2012 Physics Research Building (PRB)

Office Hours

Friday 2:00–3:00 pm in PRB 2015

Lecture Logistics

- **Location:** Scott Lab N050
- **Time:** Wed. Fri. 11:10 am – 12:30 pm
- **Schedule:** See Google Spreadsheet

Teaching Assistant & Guided Group Work

- **Graduate Teaching Assistant (GTA):** Mr. Hyungjun Jeon (jeon.258@osu.edu)
- **GTA Office Hours:** Thursday 3:00–4:00 pm in PRB 2015
- **Guided Group Work (GGW) Sessions:** Thursdays 9:30 am – 11:00 am in PRB 0118. For questions regarding GGW sessions, please contact Dr. Beatriz E Burrola Gabilondo (burrolagabilondo.1@osu.edu).

Prerequisites

The course will assume physics and mathematics knowledge at the advanced undergraduate level (e.g., basic quantum mechanics, linear algebra, and differential equations), which we will review as needed.

Lecture Notes

Lecture notes will be uploaded to Carmen as PDF files generated from LaTeX sources at least 4 days before each lecture.

Main Textbooks (Recommended but Not Required)

- *Modern Quantum Mechanics* by J.J. Sakurai and Jim Napolitano (3rd Edition, Cambridge University Press, 2020)
- *Quantum Mechanics* (Vol. 3, Lectures on Theoretical Physics) by David Tong (Cambridge University Press, 2025). Problem sets available at https://www.damtp.cam.ac.uk/user/tong/books
- *Principles of Quantum Mechanics* by Ramamurti Shankar (2nd Edition, Plenum Press, 2011)
- *Quantum Computation and Quantum Information* by Michael Nielsen and Isaac Chuang (Cambridge University Press, 2011)

Other References

- *Lectures on Quantum Mechanics and on Topics in Quantum Mechanics*, David Tong, University of Cambridge
- *Quantum Mechanics: Non-Relativistic Theory* by Lev Landau and L. Lifshitz (2nd Edition, Pergamon Press)
- *Lectures on Quantum Mechanics* by Steven Weinberg (Cambridge University Press, 2013)
- *Quantum theory: a foundational approach* by Charis Anastopoulos (Cambridge University Press, 2024)
- *Quantum Mechanics* by Arjun Berera and Luigi Del Debbio (Cambridge University Press, 2021)

Reading Assignments

For each lecture, you will be expected to complete reading assignments before class. These will be announced on Carmen. On days when reading assignments are discussed in class, you must answer questions about the reading using Carmen "Quizzes" by 10:00 am the day of class. Your answers will not be graded for correctness; full credit is given for submitting on time. Late submissions will not be counted.

Homework

Homework will be assigned weekly on Carmen (except during holidays or exams). Submissions are due on Carmen by 11:59 pm on Fridays (or occasionally Wednesdays).

HW Quizzes

To verify individual effort in solving homework problems, a 10-minute quiz will take place during the last 10 minutes (12:20–12:30 pm) of lecture on homework due dates. Handwritten answers will be scanned and uploaded to Carmen/Gradescope.

Grading Scheme

All homework, quizzes, and exams will be graded by the GTA. Refer to Table 1 for the grading allocation.

Table 1: Grading basis for the course.

Item	Time	Assignments	Percentage
Reading Quizzes	10:00 am on lecture day	23	6%
Homework	Friday at 11:59 pm	11	18%
HW Quizzes	Last 10 mins of Friday lecture	11	18%
Mid Term Exam	Oct. 9 (Fri.), 11:30 am–1:30 pm	1	29%
Final Exam	Dec. 17 (Thu.), 12:00–2:00 pm	1	29%

Exams

- **Mid Term Exam:** In-person exam scheduled for 11:30 am–1:30 pm on Friday, Oct. 9 in PRB 1080 (Smith Seminar Room).
- **Final Exam:** In-person exam scheduled for 12:00–2:00 pm on Thursday, Dec. 17 in PRB 1080 (Smith Seminar Room).

Course Topics

- Linear algebra: mathematics of quantum mechanics
- Postulates of quantum mechanics

- Quantum entanglement
- Schrödinger equation and quantum dynamics
- Heisenberg's uncertainty principle
- One-dimensional potentials
- The harmonic oscillator and coherent states
- Squeezed states and quantum optics
- Symmetry and conservation laws
- Rotational symmetry and angular momentum
- Central potentials and the hydrogen atom
- Variational methods and WKB approximation
- Perturbation theory: time-independent and time-dependent
- Scattering theory

Shared Values

Please note The Ohio State University Shared Values.

Academic Misconduct

Academic integrity is essential to maintaining an environment that fosters excellence in teaching, research, and other educational activities. The Ohio State University and the Committee on Academic Misconduct (COAM, https://oaa.osu.edu/resources/policies-and-procedures/committee-academic-misconduct) expect that all students understand the University's Code of Student Conduct (https://trustees.osu.edu/bylaws-and-rules/code). Any activity that compromises academic integrity or subverts the educational process will be reported to COAM.

Artificial Intelligence and Academic Integrity

Generative AI tools are permitted as learning aids to support your understanding (*e.g.*, asking for alternative explanations or math reviews). However, every calculation, derivation, and explanation submitted must reflect your own understanding. You may not copy or lightly modify AI-generated solutions. AI tools are strictly prohibited during timed assessments, exams, and quizzes. When AI is permitted and used for homework, include a brief acknowledgment detailing the tool and how you verified its output.

Religious Accommodations

Ohio State accommodates students' religious beliefs and practices. Requests for accommodations must be communicated in writing to the instructor within the first 14 days of the semester. See the official policy at OAA Religious Holidays Policy.

Disability Statement & Illness

Students requesting disability accommodations should register with Student Life Disability Services (https://slds.osu.edu/) and notify the instructor as early as possible. If you are ill and need to miss class, notify the instructor immediately.

Intellectual Diversity

Ohio State is committed to open inquiry and intellectual diversity. Classroom discussions aim to foster critical thinking on course material without promoting any single point of view.

Grievances and Problem Solving

To resolve grievances concerning grades or academic practices, contact the instructor first, followed by the department chairperson, college dean, and provost (Faculty Rule 3335-8-23).

Creating an Environment Free from Harassment and Discrimination

The Ohio State University does not tolerate harassment, discrimination, or sexual misconduct. Reports can be submitted to the Civil Rights Compliance Office (CRCO) at http://civilrights.osu.edu/, by phone at 614-247-5838, or via email at civilrights@osu.edu.

1 Postulates and Mathematics of Quantum Mechanics

1.1 Key experiments that established quantum mechanics (8/26)

- Refer to Weinberg Chap 1. For further reading, try Anastopoulos Chap 2, Shankar 3.

Quantum mechanics was first motivated by a series of experiments that cannot be explained by classical mechanics and statistical physics. These exceptions to classical physics happen in the microscopic world, when the scale of physical quantities, such as energy, angular momentum and electric charge becomes comparable to certain characteristic scales.

The quantization of energy was observed in both light and atomic spectra, which motivated Planck, Einstein and others to propose a discretized ("quantized") energy levels for photons and atoms. Bohr was able to explain the Rydberg formula[1] of hydrogren atom spectral lines under the assumption of a quantized angular momentum for electrons, which was later observed in the Stern-Gerlach experiment[2]. After de Broglie first proposed the wave nature of microscopic particles such as electrons, Davisson and Germer[3] were able to directly observe it in the diffraction pattern of electrons scattered by a nickel crystal. These experiments call for a new theory framework that defeats the intuition of classical physics.

1.1.1 Energy quantization of light in terms of photons

Young's double-slit experiment clearly demonstrated the wave nature of light, whose energy should change continuously in classical electromagnetism. However, this point of view is seriously challenged by black-body radiation.

1.1.1.1 Black-body radiation and Planck's formula

Consider light in a cubic box (a simplified model of a black body) of volume L^3. A standing wave in the box satisfies the following condition for its wavevector $\vec{k} = (k_x, k_y, k_z)$:

$$\lambda_i = \frac{2\pi}{k_i} = \frac{2L}{n_i}, \quad n_i = 1, 2, 3, \cdots, \quad i = x, y, z \tag{1.1}$$

Counting the number of standing wave modes in the box in terms of frequency $\nu = ck/2\pi$

$$\int \mathrm{d}k_x \mathrm{d}k_y \mathrm{d}k_z = \frac{1}{8} 4\pi \int k^2 \mathrm{d}k = (\frac{\pi}{L})^3 \sum_{\{n_i\}} = (\frac{\pi}{L})^3 N_{\text{modes}} \tag{1.2}$$

we have the density of states for light-wave modes per unit volume:

$$\frac{1}{L^3}\frac{\mathrm{d}N}{\mathrm{d}\nu} = (\frac{1}{\pi})^3 \frac{\pi}{2} k^2 \frac{\mathrm{d}k}{\mathrm{d}\nu} = \frac{8\pi}{c^3}\nu^2 \tag{1.3}$$

Assuming the equipartition theorem in classical physics, i.e. each mode carries an averaged energy of $k_B T$, the black-body radiation energy per unit volume is

$$\frac{\mathrm{d}E}{L^3 \mathrm{d}\nu} = \frac{k_B T}{L^3}\frac{\mathrm{d}N}{\mathrm{d}\nu} = \frac{8\pi k_B T}{c^3}\nu^2 \tag{1.4}$$

This is known as the Rayleigh-Jeans formula and it correctly captures the long-wavelength (small ν) behavior of universal black-body radiation, first established by Kirchhoff around 1860. However, it diverges in the short wavelength limit, known as the ultraviolet catastrophe. Experimentally the Wien's Distribution Law was observed at short wavelengths:

$$\frac{\mathrm{d}E}{L^3 \mathrm{d}\nu} \sim \nu^3 e^{-\alpha(T)\nu} \tag{1.5}$$

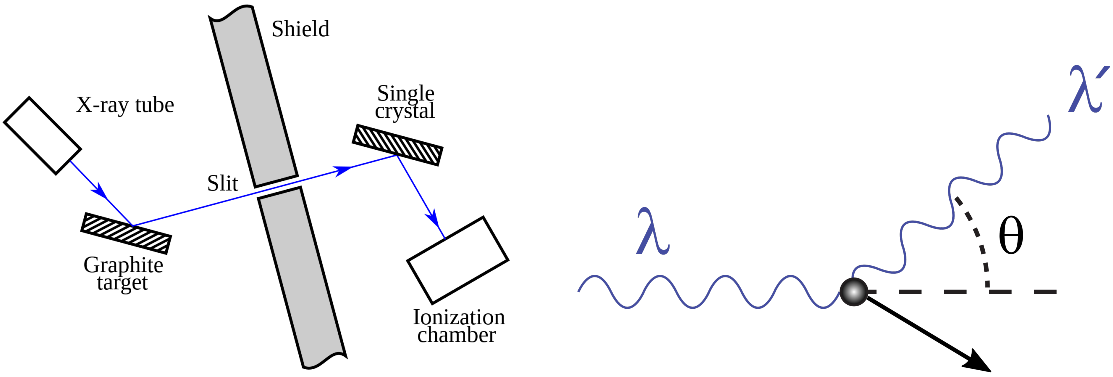

Figure 1.1: (Left) Schematic setup of Compton's experiment. (Right) The light-electron scattering process in the Compton experiment.

In order to reconcile long-wavelength Rayleigh-Jeans formula to short-wavelength Wien's law, Planck came up with his famous formula:

$$\frac{\mathrm{d}E}{L^3\mathrm{d}\nu} = \frac{8\pi\nu^2}{c^3}\frac{h\nu}{e^{\alpha\nu}-1}, \quad \alpha = \frac{h}{k_B T} \tag{1.6}$$

where the Planck constant is

$$h = 6.63 \times 10^{-34}\ J \cdot s \tag{1.7}$$

Planck's formula (1.6) indicates that the statistical-averaged energy of light at frequency ν is given by

$$U(\nu) = \frac{h\nu}{e^{h\nu/K_B T}-1} = \frac{\sum_{n\in\mathbb{Z}_+} nh\nu e^{-nh\nu/K_B T}}{\sum_{n\in\mathbb{Z}_+} e^{-nh\nu/k_B T}} \tag{1.8}$$

under Planck's quantization hypothesis that the microscopic states of light have energy $nh\nu$ where $n \in \mathbb{Z}_+$ is a positive integer.

1.1.1.2 Einstein's "light quantum" and photoelectric effect

Motivated by Planck's quantization hypothesis, based on the assumption that light gives out energy in multiples of energy quantum $h\nu$, Einstein predicted the photoelectric effect

$$K_{\max} = eV = h\nu - \Phi \tag{1.9}$$

where K_{max} is the maximal kinetic energy of emitted photoelectrons when light of frequency ν shines on a metal, and Φ is the work function of the metal. V is minimal voltage that is required to stop photoelectron emission. This experiment was carried out by Millikan[4], who obtained Planck's constant

$$h = e\frac{\mathrm{d}V}{\mathrm{d}\nu} \tag{1.10}$$

1.1.1.3 Compton effect demonstrated quantization of light in terms of photons

Discovered by Arthur Compton in 1923[5], when the energy of the X ray photon (~17 keV) was significantly larger than the binding energy of the atomic electron, the electrons could be treated as being free after scattering. In this case, a *Compton shift* between the wavelengths of injected light (λ) and the scattered light (λ') was observed to satisfy the following relation with the scattering angle θ (see Fig. 1.1):

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta) \tag{1.11}$$

where h is the Planck's constant.

The Compton shift can be explained by energy and momentum conservation, by assuming the photon energy of $E = h\nu = hc/\lambda$.

1.1.2 Quantization of atomic energy levels

1.1.2.1 Rydberg formula for the spectral lines in a hydrogen atom

After the discovery of dark lines in the spectrum of the sun at the beginning of 18th century, it was later realized that hot atomic gases absorb and emit light only at certain definite frequencies. Building on works of Balmer and others, Rydberg discovered the following rule[1] for the frequency of emitted light from hydrogen gases:

$$\nu = \frac{c}{\lambda} = cR_H(\frac{1}{n^2} - \frac{1}{m^2}), \quad m, n \in \mathbb{Z}_+. \tag{1.12}$$

This clearly demonstrated the discreteness of atomic energy levels $E_n \propto \frac{1}{n^2}$, further motivating Bohr's theory of the hydrogen atom.

1.1.2.2 Franck-Hertz experiment

In the Franck-Hertz experiment (1914) illustrated in Fig. 1.2, in a vacuum tube fitted with three electrodes (an electron-emitting, hot cathode; a metal mesh grid; and an anode), when a grid voltage is applied between the cathode and the anode, the electric current through the tube filled with thin mecury vapours, due to electrons that pass through the grid and reach the anode, is measured in the experiment. As shown in the middle figure, the current increases rapidly at low voltages, and then sharply drops almost back to zero at 4.9 V, with similar effects observed as 9.8 V and other multiples of 4.9 V. This can be understood by the quantized energy levels of mercury atoms. inside the single mercury droplet in the vacuum tube, as shown in Fig. 1.2.

1.1.3 Quantization of angular momentum

1.1.3.1 Bohr's explanation of hydrogen atom spectral lines

Rydberg's rule[1] indicates that the photon energy emitted by the hydrogen atom is given by

$$h\nu = E_m - E_n, \quad E_n = -\frac{hcR_H}{n^2}, \; n \in \mathbb{Z}_+. \tag{1.13}$$

How to explain the $1/n^2$ quantization of hydrogen atom energy levels? According to classical mechanics, for an electron in the Coulomb force field of hydrogen nucleus, its speed v and radius r satisfies

$$\frac{mv^2}{r} = \frac{ke^2}{r^2} \Longrightarrow E = \frac{1}{2}mv^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r} \tag{1.14}$$

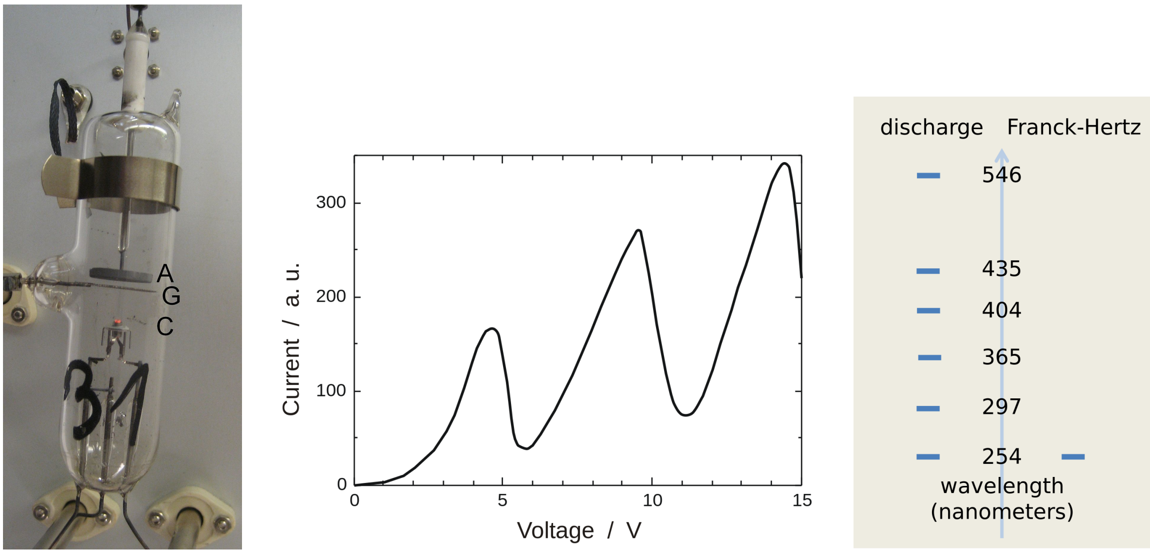

Figure 1.2: (Left) Photograph of a vacuum tube used for the Franck-Hertz experiment. (Middle) Anode current (arbitrary units) versus the grid voltage relative to the cathode. (Right) Wavelengths of light emitted by a mercury vapor discharge and by a Franck-Hertz tube in operation at 10 V. (Picture from Wikipedia)

where k is the Coulomb constant. To fit the n^2 dependence of the orbit radius r, Niels Bohr came up with the following quantization condition for angular momentum:

$$mvr = n\hbar \tag{1.15}$$

where $\hbar$ turns out to be $h/2\pi$. This quantization condition leads to the famous energy levels of electrons in a hydrogen atom:

$$E_n = -\frac{mk^2e^4}{2\hbar^2}\frac{1}{n^2} = -\frac{13.6\text{ eV}}{n^2} \tag{1.16}$$

1.1.3.2 Stern-Gerlach experiment demonstrated the quantization of angular momentum

In 1922 Stern and Gerlach performed an experiment[2] measuring the deflection of silver atoms by an inhomogeneous magnetic field, as illustrated in Fig. 1.3. Due to the interaction energy $U = -\vec{\mu}\cdot\vec{B}$ of a magnetic moment $\vec{\mu}$ under a magnetic field $\vec{B} = B_z\hat{z}$, a non-uniform magnetic field exerts a force $\vec{F} = -\vec{\nabla}U = \mu_z\vec{\nabla}B_z$ on the moment-carrying silver atoms, therefore deflecting the silver atom beams in the inhomogenous magnetic field. As shown in Fig. 1.3, in contrast a continuous range of values of the magnetic moment as expected in classical mechanics, the experimental observations pointed to quantization of magnetic moment $\mu_z = g\mu_B S_z/\hbar$, where we defined μ_B as the Bohr magneton:

$$\mu_B = \frac{\hbar e}{2m_e} \tag{1.17}$$

$g \approx 2$ as the electron g-factor and

$$S_z = \pm\frac{\hbar}{2} \tag{1.18}$$

is the quantized spin angular momentum of a silver atom observed in Stern-Gerlach experiment.

This experiment demonstrated that in addition to energy level quantization, other physical observable such as spin angular momentum are also quantized in microscopic scales such as in

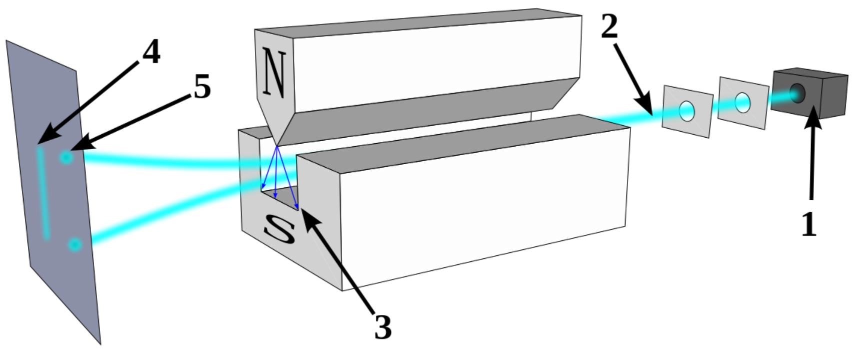

Figure 1.3: Schematic setup of Stern-Gerlach experiment: Silver atoms travelling through an inhomogeneous magnetic field, being deflected up or down depending on their spin. (1) furnace, (2) beam of silver atoms, (3) inhomogeneous magnetic field, (4) the classically expected result, (5) the observed result.

silver atoms. Moreover, we can consider sequential Stern-Gerlach experiments shown in Fig. 1.4, where silver atoms are deflected by multiple inhomogeneous magnetic fields in a sequence. Remarkably, unlike the classical case of light passing through polarizers, which filter out everything after two polarizers with perpendicular polarization axes, two quantum mechanical "beam splitters" with perpendicular magnetic field directions still allows two splitted beams to pass through, as illustrated in the middle and bottom parts of Fig. 1.4.

1.1.4 Davisson-Germer experiment demonstrated the wave nature of electrons

In 1924 de Broglie formally proposed the wave nature of electrons and other particles, in terms of their de Broglie wavelengths:

$$\lambda = \frac{h}{p} \tag{1.19}$$

where p is the momentum of the particle. If this proposal were true, it means that microscopic particles such as electrons should exhibit interference phenomena in the same way as acoustic and light waves.

In 1927 Davisson and Germer[3] performed an experiment to observe the diffraction pattern of electrons scattered by the surface of a crystal of nickel metal, as shown in Fig. 1.5. The highest intensity was observed at an angle $\theta = 50^o$ with a voltage of 54 V, giving the electrons a kinetic energy of 54 eV. To be compatible with the Bragg's law

$$n\lambda = 2d\sin(90^o - \theta/2), \quad n \in \mathbb{Z}_+. \tag{1.20}$$

with the lattice spacing of nickel $d = 0.091$ nm, the $n = 1$ Bragg peak in (1.20) exactly corresponds to the de Broglie wavelength (1.19) of electrons. We will examine the experimental

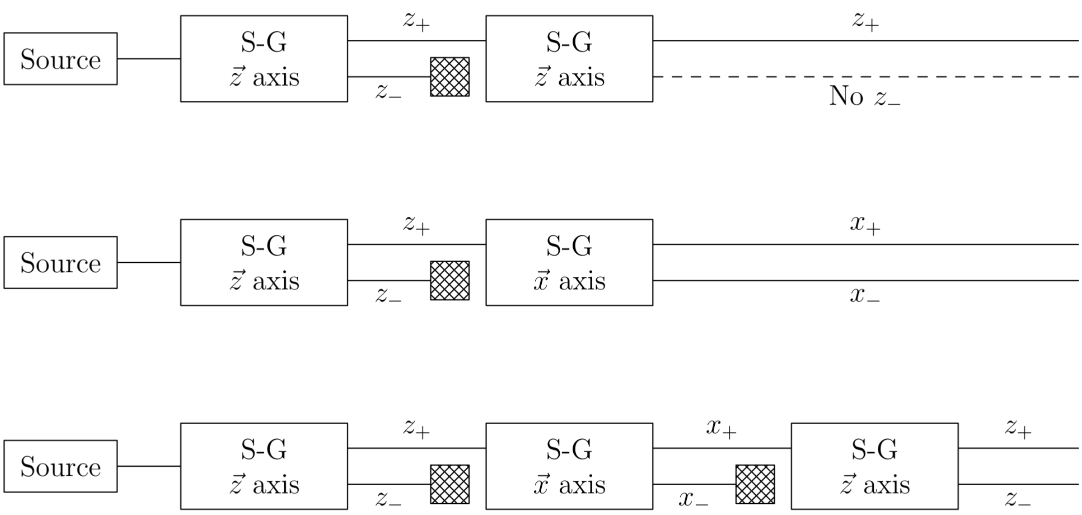

Figure 1.4: Sequential Stern-Gerlach experiments. (Picture from S&N)

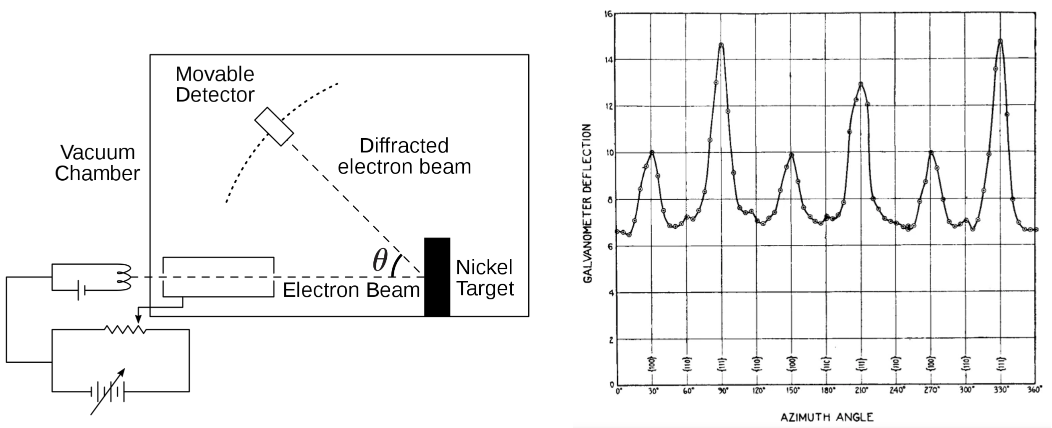

Figure 1.5: (Left) Schematic setup of Davisson-Germer experiment. (Right) Intensity of electron scattering vs. azimuth angle at 54 volts, co-latitude $\theta = 50^o$.

results in Fig. 1.5 more carefully in Homework 1.

1.2 Hilbert space and state vectors (8/28)

- Refer to N&C 2.1, Tong 3.1, Shankar 1.1-1.4

How to explain the above experimental observations? 4 postulates, derived (or more precisely, guessed) after a long process of trial and error, form the foundation for a theory framework that explains all observed experimental phenomena, known as quantum mechanics. Below we will introduce these postulates one by one, along the way of introducing linear algebra as the mathematics of quantum mechanics.

Postulate 1.1. *Associated to any isolated physical system is a complex vector space with inner product (i.e. a Hilbert space) known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the system's state space.*

The key mathematical framework for quantum mechanics is linear algebra. The basic objects of linear algebra are vector spaces:

A *vector space* over a *scalar field* $\mathbb{F}$ is a set $\mathbb{V}$ of objects (called *vectors*) equipped two binary operations: a *vector addition* ("$+''$)

$$|v\rangle + |w\rangle \in \mathbb{V}, \quad \forall\ |v\rangle, |w\rangle \in \mathbb{V}. \tag{1.21}$$

and a *scalar multiplication*:

$$a|v\rangle \in \mathbb{V}, \quad \forall\ |v\rangle \in \mathbb{V},\ a \in \mathbb{F}. \tag{1.22}$$

Mathematically, the set $\mathbb{V}$ forms an abelian group under vector addition operation and the scalar multiplication defines a ring homomorphism from the field $\mathbb{F}$ into the endomorphism ring of this abelian group. We will break down these terminology below.

It is worthwhile to define a group since we will study symmetry in quantum mechanics later. A group is a set $V = \{u, v, \cdots\}$ equipped with a *closed* binary operation $+$: $V \times V \to V$

$$u + v = w \in V, \quad \forall\ u, v \in V. \tag{1.23}$$

where 3 group axioms are satisfied:

(1) *Identity element*:

$$\exists\ 0 \in V,\ \text{s.t.}\ \forall\ u \in V,\ u + 0 = 0 + u = u. \tag{1.24}$$

(2) *Inverse*:

$$\forall\ u \in V,\ \exists\ (-u) \in V,\ \text{s.t.}\ u + (-u) = 0. \tag{1.25}$$

(3) *Associativity*:

$$(u + v) + w = u + (v + w), \quad \forall\ u, v, w \in V. \tag{1.26}$$

An Abelian group is a group whose addition operation is commutative:

$$u + v = v + u, \quad \forall\ u, v \in V. \tag{1.27}$$

The scalar multiplication is another map $\mathbb{F} \times \mathbb{V} \to \mathbb{V}$ that is *associative*

$$a(b|v\rangle) = (ab)|v\rangle, \quad \forall\ a, b \in \mathbb{F},\ |v\rangle \in \mathbb{V} \tag{1.28}$$

and *distributive in both the vectors and the scalars*:

$$a(|u\rangle + |v\rangle) = a|u\rangle + a|v\rangle, \tag{1.29}$$
$$(a+b)|v\rangle = a|v\rangle + b|v\rangle, \quad \forall\ a, b \in \mathbb{F},\ |u\rangle, |v\rangle \in \mathbb{F}. \tag{1.30}$$

The above conditions satisfied by vector addition and scalar multiplication are summarized well in Chapter 1.1 of Shankar.

Of particular interests to quantum mechanics is a *Hilbert space*, which is a (n-dimensional) vector space $\mathbb{C}^n$ over the field $\mathbb{C}$ of complex numbers, equipped with a map $\mathbb{V} \times \mathbb{V} \to \mathbb{C}$ known as an *inner product*:

$$\langle u|v\rangle \in \mathbb{C}, \quad \forall\ |u\rangle, |v\rangle \in \mathbb{V}. \tag{1.31}$$

In the Dirac notation, $\langle u|$ is known as a *bra*, and $|v\rangle$ is known as a *ket*. The inner product is *skew-symmetric*

$$\langle u|v\rangle = \langle v|u\rangle^* \tag{1.32}$$

positive semidefinite

$$\langle v|v\rangle \geq 0, \quad \langle v|v\rangle = 0 \Longleftrightarrow |v\rangle = 0. \tag{1.33}$$

and *linear in ket*

$$\langle u|(\sum_i \lambda_i |v_i\rangle) = \sum_i \lambda_i \langle u|v_i\rangle \tag{1.34}$$

These properties are summarized in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C.

Note that an inner product also defines a norm for each vector in vector space $\mathbb{V}$:

$$||v\rangle| \equiv \sqrt{\langle v|v\rangle} \tag{1.35}$$

Two important properties of an inner product are

Theorem 1.2. ***Cauchy-Schwarz inequality:***

$$\langle v|v\rangle\langle u|u\rangle \geq |\langle u|v\rangle|^2 \tag{1.36}$$

The equality holds if and only if $|v\rangle$ and $|w\rangle$ are linearly dependent.

Theorem 1.3. ***Triangle inequality*** *for the norms of vectors:*

$$||v\rangle + |w\rangle| \leq ||v\rangle| + ||w\rangle| \tag{1.37}$$

Triangle inequality can be derived from the Cauchy-Schwarz inequality, which is a part of Homework 2.

Strictly speaking, in addition to being a vector space equipped with an inner product (known as an inner product space), a Hilbert space must also be *complete* in the sense that every Cauchy sequence $\{v^{(k)}\}$ of vectors in $\mathbb{V}$ has a limit that also lies in $\mathbb{V}$:

$$\lim_{k\to\infty} v^{(k)} \in \mathbb{V}, \quad \text{if } \forall\ \epsilon > 0, \quad \exists\ N \in \mathbb{Z}_+,\ \text{s.t. } |v^{(m)} - v^{(n)}| < \epsilon, \ \ \forall\ m, n > N. \tag{1.38}$$

For an inner product space of finite dimensions on any complete field $\mathbb{F}$, such as real and complex numbers $\mathbb{F} = \mathbb{R}, \mathbb{C}$, it is automatically a Hilbert space. Below are a few examples of Hilbert spaces, and inner product spaces that are not Hilbert spaces:

- n-dimensional Hilbert space on complex numbers: $\mathbb{V} \simeq \mathbb{C}^n$
- $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$ (see Homework 1)
- n-dimensional Hilbert space on real numbers: $\mathbb{V} \simeq \mathbb{R}^n$
- n-dimensional vector space $\mathbb{V} \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.
- The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1|f_2\rangle \equiv \int_a^b f_1^*(x) f_2(x) \mathrm{d}x \tag{1.39}$$

  is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.
- For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

A Hilbert space forms the *state space* described in Postulate 1.1. With the help of the inner product structure, we can always choose a complete set of orthonormal basis $\{|i\rangle\}$ that expands the full state space $\mathbb{V}$, such that

$$|v\rangle = \sum_i v_i |i\rangle, \ \forall \ |v\rangle \in \mathbb{V}; \quad \langle i|j\rangle = \delta_{i,j}, \quad v_i = \langle i|v\rangle. \tag{1.40}$$

where $\delta_{i,j}$ is the Kronecker delta function. Given a n-dimensional vector space V, one can always find a set of n linearly independent vectors $\{e_j | 1 \le j \le n\}$, known as a basis of vector space V, such that any vector $v \in V$ can be uniquely expressed as a linear superposition of the basis vectors: $v = \sum_j v_j e_j$ with $v_j \in \mathbb{F}$. Given the inner product structure in a Hilbert space, a set of complete orthonormal basis can be constructed following the Gram-Schmidt procedure, discussed in details in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C. In such a complete orthonormal basis of Hilbert space $\mathbb{V} = \mathbb{C}^n$, a Dirac ket $|v\rangle$ can be represented by a column vector $\vec{v} = (v_1, \cdots, v_n)^T$, a Dirac bra represented by a row vector, and the inner product reduces to the familiar form in $\mathbb{C}^n$:

$$\langle u|v\rangle = \sum_i u_i^* v_i = \left(u_1^*, \cdots, u_n^*\right) \begin{pmatrix} v_1 \\ \cdots \\ v_n \end{pmatrix} \tag{1.41}$$

The column vector $(v_1, \cdots, v_n)$ is known as the *wavefunction* of state $|v\rangle$ in the basis set $\{|i\rangle\}$.

A special set of complete orthonormal basis that plays a particularly important role in quantum mechanics is the eigenstates $|x\rangle$ of position operator $\hat{x}$, where we consider one dimension for simplicity. For any normalized state $|\psi\rangle$, its continuous-variable wavefunction

$$\psi(x) \equiv \langle x|\psi\rangle, \qquad \int \mathrm{d}x |\psi(x)|^2 = \langle\psi|\psi\rangle = 1. \tag{1.42}$$

determines the probability of detecting the quantum-mechanical particle at position x:

$$\Pr(x) = |\psi(x)|^2, \qquad \int \mathrm{d}x \Pr(x) = 1. \tag{1.43}$$

1.3 Linear operators and observables (9/2)

- Refer to N&C 2.1; Tong 3.2, 4.1; S&N 1.3-1.4; Shankar 1.5-1.6

Given two vector spaces $\mathbb{V}$ and $\mathbb{W}$, we can also define a linear operator $\hat{O}$ as any function that maps the vector space $\mathbb{V}$ to $\mathbb{W}$, which is linear in its inputs:

$$\hat{O}(\sum_i a_i|v_i\rangle) = \sum_i a_i\hat{O}(|v_i\rangle) \tag{1.44}$$

In the case of $\mathbb{V} = \mathbb{W}$ we call it a linear operator $\hat{O} \in \mathbb{L}_\mathbb{V}$ on vector space $\mathbb{V}$, which is of particular interest to quantum mechanics.

Using a complete set of orthonormal basis $\{|j\rangle|1 \leq j \leq \dim\mathbb{V}\}$ for Hilbert space $\mathbb{V}$, each linear operator has a matrix representation:

$$\hat{O} = \sum_{a,b} O_{a,b}|a\rangle\langle b|, \quad O_{a,b} \equiv \langle a|\hat{O}|b\rangle \tag{1.45}$$

In particular, for the identity matrix which maps every vector to itself, we have the **completeness relation** for basis $\{|i\rangle\}$, also known as the **resolution of identity**:

$$\sum_i |i\rangle\langle i| = \hat{1}. \tag{1.46}$$

In the case of continuous variables, e.g. coordinates for a particle in free space, the summation will be replaced by integral:

$$\int \mathrm{d}x|x\rangle\langle x| = \hat{1} \tag{1.47}$$

where $|x\rangle$ is an eigenstate of position operator $|x\rangle$ with eigenvalue x.

We can also define the product of two linear operators $\hat{M}, \hat{N} \in \mathbb{L}_\mathbb{V}$:

$$\hat{M}\hat{N} = \sum_{a,b,c} M_{a,b}N_{b,c}|a\rangle\langle c| \tag{1.48}$$

their commutator

$$[\hat{M}, \hat{N}] = \hat{M}\hat{N} - \hat{N}\hat{M} \tag{1.49}$$

and anti-commutator

$$\{\hat{M}, \hat{N}\} = \hat{M}\hat{N} + \hat{N}\hat{M} \tag{1.50}$$

and the *Hermitian conjugate* (or *Hermitian adjoint*) of an operator

$$\hat{O}^\dagger = \sum_{a,b} O^*_{b,a}|a\rangle\langle b| = (O^T)^* = (O^*)^T \tag{1.51}$$

Specifically in Postulate 1.9 about quantum (projective) measurements, each physical observable corresponds to a Hermitian linear operator, represented as a Hermitian matrix in the orthonormal basis:

$$\hat{M} = \hat{M}^\dagger = \sum_{a,b} M_{a,b}|a\rangle\langle b|, \quad (M^\dagger)_{a,b} = M^*_{b,a} = M_{a,b} \equiv \langle a|\hat{M}|b\rangle. \tag{1.52}$$

The diagonal form

$$\hat{O} = \sum_a O_a |a\rangle\langle a| \tag{1.53}$$

where $\{|a\rangle\}$ is a complete set of orthonormal basis of the Hilbert space, is also known as the *spectral decomposition* of a matrix $\hat{O}$. If there is a complete set of orthonormal basis $\{|a\rangle\}$ such that the linear operator $\hat{O}$ can be rewritten as the diagonal form (1.53), the matrix $\hat{O}$ is called "diagonalizable".

Not all matrices are diagonalizable (i.e. have a diagonal form), two non-diagonalizable examples are

$$\hat{M}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \; \hat{M}_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \tag{1.54}$$

$\hat{M}_1$ is known as a Jordan block with only 1 eigenvector, while $\hat{M}_2$ has two eigenvectors with different eigenvalues, but not orthogonal to each other. Generally, any complex square matrix can be decomposed into tensor sums of Jordan blocks by similarity transformations.

Below are a few useful theorems on the diagonalization of matrices:

Theorem 1.4. ***The spectral decomposition theorem***: *A linear operator $\hat{O}$ on vector space $\mathbb{V}$ is diagonal w.r.t. some orthonormal basis of $\mathbb{V}$ if and only if operator $\hat{O}$ is* normal *i.e. $\hat{O}^\dagger \hat{O} = \hat{O}\hat{O}^\dagger$.*

Refer to N&C Box 2.2 for proof of this theorem.

Theorem 1.5. *A normal matrix is Hermitian if and only if it has real eigenvalues.*

Theorem 1.6. *The eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal.*

Theorem 1.7. *A positive linear operator must be Hermitian.*

Theorem 1.8. ***Simultaneous diagonalization theorem***: *Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagnoalized simultaneously if and only if $[\hat{A}, \hat{B}] = 0$.*

In other words, there is a complete set of simultaneous eigenket $\{|a, b\rangle\}$ such that

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \tag{1.55}$$

if and only if $[\hat{A}, \hat{B}] = 0$.

An Hermitian operator can be diagonalized into a special orthonormal basis $\{|i\rangle\}$, known as its eigenbasis, in which the linear operator is represented by a diagonal matrix

$$\hat{M} = \sum_i \lambda_i |i\rangle\langle i| \tag{1.56}$$

If we use the set $\{m\}$ to label all different (real) eigenvalues of matrix $\hat{M}$, we can rewrite the above diagonal form (1.56) as

$$\hat{M} = \sum_m m\hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle\langle j| \tag{1.57}$$

where operators $\hat{P}_m$ are known as *projection operators* (or *projectors*) onto the eigenspace of $\hat{M}$ with eigenvalue m. Mathematically, a projector $\hat{P}$ is defined as a linear operator satisfying:

$$\hat{P}^2 = \hat{P} \tag{1.58}$$

You will learn more about projectors in HW2.

Given a state $|\psi\rangle$ and a physical observable $\hat{M}$, the (probability-weighted) *expectation value* $\langle\hat{M}\rangle$ of operator $\hat{M}$ w.r.t. state vector $|\psi\rangle$ is

$$\langle\hat{M}\rangle \equiv \frac{\langle\psi|\hat{M}|\psi\rangle}{\langle\psi|\psi\rangle} \tag{1.59}$$

Naturally, the expectation value of any physical observable ia a real number, since any physical observable corresponds to a Hermitian operator.

We note that the expectation value (1.59) of any observable is the same for two linearly dependent states $|\psi\rangle$ and $\alpha|\psi\rangle$ for any complex scalar $\alpha \in \mathbb{C}$. This means $|\psi\rangle$ and $\alpha|\psi\rangle$ describe the same physical state. In other words, in a N-dimensional Hilbert space $\mathbb{V} = \mathbb{C}^N$, for a generic state

$$|\psi\rangle \equiv \sum_{j=1}^{N} \psi_j |j\rangle \tag{1.60}$$

where $\{|j\rangle | 1 \leq j \leq N\}$ is a complete set of orthonormal basis of $\mathbb{V}$, although $\{\psi_j \in \mathbb{C} | 1 \leq j \leq N\}$ correspond to $2N$ independent real variables, only $2N-2$ of them are physically measurable, because the global phase factor and normalization leads to 2 redundancies that cannot be physically observed.

In the simplest example of a single qubit, we have $\mathbb{V} = \mathbb{C}^2$ for the $N = 2$-dimensional Hilbert space. There are 4 linearly independent Hermitian operators in $\mathbb{C}^2$, whose matrix representations are

$$\hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.61}$$

As you will learn in HW1, they (up to normalization) form a complete set of orthonormal basis for the Hilbert space $L_{\mathbb{C}^2}$ of linear operators. In the matrix representation, a generic state vector can be written as

$$|\psi\rangle = \alpha \begin{pmatrix} \cos\theta e^{\mathrm{i}\phi} \\ \sin\theta \end{pmatrix}, \quad \alpha \in \mathbb{C} \tag{1.62}$$

θ and ϕ are the two variables in the state vector that can be physically measured. One can carry out measurements for two observables to extract θ and ϕ:

$$\langle\hat{\sigma}_z\rangle = \cos(2\theta), \quad \langle\hat{\sigma}_x\rangle = \sin(2\theta)\cos\phi \tag{1.63}$$

1.3.1 Appendix: Functions of operators/matrices

A function $f(\hat{M})$ of a square matrix $\hat{M}$ can be defined by the following Taylor expansion

$$f(\hat{M}) = \sum_{n=0}^{\infty} \frac{\hat{M}^n}{n!} \frac{\mathrm{d}^n f(x)}{\mathrm{x}^n}|_{x=0} \tag{1.64}$$

if the Taylor expansion converges at $x = 0$. In quantum mechanics we will mostly encounter square matrices that can be diagonalized, such as Hermitian and unitary matrices.

For a diagonalizable matrix $\hat{M}$ with eigenkets $\{|j\rangle\}$ and eigenvalues $\{\lambda_j\}$, the matrix can be written as

$$\hat{M} = \sum_j \lambda_j |j\rangle\langle j| \tag{1.65}$$

and a function of this diagonalizable matrix can be defined as

$$f(\hat{M}) = \sum_j f(\lambda_j) |j\rangle\langle j| \tag{1.66}$$

1.4 Projective measurements (9/4)

- Refer to N&C 2.2.4-2.2.7, Tong 3.4, 4.2, 16.4.1

Postulate 1.9. *A* ***projective measurements*** *is described by a physical observable, $\hat{M} = \hat{M}^\dagger$ on the Hilbert space of the system being observsed. The observable has a specral decomposition*

$$\hat{M} = \sum_m m\hat{P}_m \tag{1.67}$$

where $\hat{P}_m$ is the projector onto the eigenspace of M with eigenvalue m. The possible outcomes of the measurement correspond to the eigenvalues m of the observable. Upon measuring state $|\psi\rangle$, the probability of getting result m is given by

$$p(m) = \langle\psi|\hat{P}_m|\psi\rangle \tag{1.68}$$

If the outcome m occured, the state of the quantum system collapsed into a new state

$$\hat{P}_m|\psi\rangle/\sqrt{p(m)} \tag{1.69}$$

immediately after the measurement.

Postulate 1.9 is in fact only a special form of quantum measurements, which will be discussed in detail in later parts of this course. Most generally, a generalized quantum measurement is described as follows:

Postulate 2a: A **generalized quantum measurement** is described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the (normalized) state of the quantum system is $|\psi\rangle$ immediately before the measurements, then the probability that result m occurs is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle \tag{1.70}$$

and the state of the system after the measurement is $\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$. The measurement operators satisfy the completeness equation

$$\sum_m M_m^\dagger M_m = \hat{1}. \tag{1.71}$$

When we choose $\hat{M}_m$ to be projector $\hat{P}_m$ into the eigenspace of observable $\hat{M}$ with eigenvalue m, the above generalized measurement reduces to the a projective measurement. Now it also becomes clear that the expectation value of observable $\hat{M}$ w.r.t. to (normalized) state vector $|\psi\rangle$ is nothing but the probability-weighted average of measurement outcomes:

$$\langle\hat{M}\rangle \equiv \langle\psi|\hat{M}|\psi\rangle = \sum_m m \times p(m) \tag{1.72}$$

One can also find out the *standard deviation* of measuring observable $\hat{M}$ in state $|\psi\rangle$:

$$\Delta(M) \equiv \sqrt{\langle(\hat{M} - \langle\hat{M}\rangle)^2\rangle} = \sqrt{\langle\hat{M}^2\rangle - \langle\hat{M}\rangle^2} \tag{1.73}$$

A significant consequence of the collapse part in Postulate 1.9 is the following: starting from any state vector $|\psi\rangle$, if we measure the same observable $\hat{M}$ multiple times, the measurement outcome does not change any more after the 1st measurement. This is because the state already collapse into an eigenstate of $\hat{M}$ with some eigenvalue m after the 1st measurement.

Another interesting aspect comes from *incompatible measurements* of physical observables that do not commute with each other. If we perform two measurements $\hat{A}$ and $\hat{B}$ that satisfy $[\hat{A}, \hat{B}] \neq 0$, for a generic quantum state, the measurement outcome depend on the order of the two measurements: i.e. measuring $\hat{A}$ after $\hat{B}$ versus $\hat{B}$ after $\hat{A}$ will yield different results. For example, starting from the eigenstate $|\uparrow\rangle$ of Pauli matrix σ_z, if we measure σ_z and then σ_x, there is a probability of $1/2$ for us to end up with $+1$ outcomes for both measurements. However, if σ_x is measured before the σ_z measurement, there is only a probability of $1/4$ of $+1$ outcomes for both measurements.

1.5 Unitary evolution of a closed quantum system (9/9)

- Refer to S&N 2.1; Tong 3.3; N&C 2.1.8, 2.2.2, 4.2; Shankar 4.3

In addition to Postulates 1.1 and 1.9 introduced earlier, we have two more postulates, one regarding how a quantum system evolves with time:

Postulate 1.10. *The time evolution of the state of a closed quantum system is described by the Schrodinger equation:*

$$\mathrm{i}\hbar\frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle \tag{1.74}$$

where $\hbar = h/2\pi \approx 1.05 \times 10^{-34}$ *J·s is a constant, and* $\hat{H}$ *is a fixed Hermitian operator known as the Hamiltonian of the closed system.*

Postulate 1.10 states that the state vector associated with a closed quantum system evolves following the Schrodinger equation (1.74). This postulate can be states in an equivalent manner:

Postulate 3a *The evolution of a closed quantum system is described by a unitary transformation. That is, the state* $|\psi(t_2)\rangle$ *of the system at time* t_2 *is related to the state* $|\psi(t_1))\rangle$ *at time* t_1 *by a unitary operator* U *which only depends on the times* t_1 *and* t_2*:*

$$|\psi(t_2)\rangle = U(t_1,t_2)|\psi(t_1)\rangle, \quad U(t_1,t_2) = e^{-\mathrm{i}(t_2-t_1)\hat{H}/\hbar} \tag{1.75}$$

One can further generalize this to a time-dependent Hamiltonian $\hat{H}(t)$ where we have

$$U(t_1,t_2) = \mathcal{T}e^{-\mathrm{i}\int_{t_1}^{t_2}\hat{H}(t)\mathrm{d}t/\hbar} \tag{1.76}$$

where $\mathcal{T}$ denotes time ordering in the integral, as explained in S&N 2.1.

In linear algebra, a unitary operator $\hat{U}$ is defined as any linear operator satisfying

$$\hat{U}^\dagger\hat{U} = \hat{1}. \tag{1.77}$$

which also implies $\hat{U}\hat{U}^\dagger = \hat{1}$. Due to the spectrum decomposition theorem, a unitary operator can be diagonalized, and its eigenvalues all have modulus 1:

$$\hat{U}|\psi\rangle = e^{\mathrm{i}\theta}|\psi\rangle \tag{1.78}$$

Three examples of unitary operators in quantum mechanics are the following: (1) unitary time evolution operator $U(t_1,t_2) = e^{-\mathrm{i}(t_2-t_1)\hat{H}/\hbar}$ for a closed quantum system discussed above; (2) a symmetry operator in a quantum system as we will discuss later in this course; (3) a quantum (logic) gate in quantum computing. For instance, in a two-dimensional closed quantum system, also known as a qubit, the common quantum gates are summarized in Table 1.1.

Starting from a complete set of orthonormal basis vectors $\{|i\rangle\}$ of Hilbert space $\mathbb{V}$, after the action of a unitary operator $\hat{U}$, we can obtain another complete set of orthonormal basis vectors $\{|i\rangle' = \hat{U}|i\rangle\}$ of the same Hilbert space. In other words, each unitary operator implements a change of basis in the Hilbert space. Therefore, the unitary time evolution merely performs a rotation for vectors in the Hilbert space, since it only changes the state itself but not the inner product of any two states.

In the case of unitary time evolution $U(t_1,t_2)$, the Hamiltonian $\hat{H}$ of a closed quantum system plays a crucial role. In its spectral decomposition:

$$\hat{H} = \sum_a E_a|a\rangle\langle a| \tag{1.79}$$

Single-qubit gate	Symbol	Unitary matrix
Pauli-X (bit flip)	X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli-Y	Y	$\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$
Pauli-Z (phase flip)	Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Hadamard	H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
Phase	S	$\begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}$
$\pi/8$	T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{pmatrix}$

Table 1.1: Common single-qubit gates and associated unitary matrices.

the eigenkets $|a\rangle$ are called the *energy eigenstates*, or *stationary states*, and the eigenvalue E_a is called the energy of state $|a\rangle$. In particular, the eigenstate with the lowest energy is known as the *ground state*, while other states with higher energy are known as *excited states*. According to Schrodinger equation (1.74), a generic state vector

$$|\psi\rangle = \sum_a c_a|a\rangle \tag{1.80}$$

evolves as

$$|\psi(t)\rangle = e^{-\mathrm{i}t\hat{H}/\hbar}|\psi\rangle = \sum_a c_a e^{-\mathrm{i}E_a t/\hbar}|a\rangle \tag{1.81}$$

and the different oscillation frequency $\omega_a = E_a/\hbar$ in different components of state $|\psi(t)\rangle$ can result in oscillations of physical observables. Two such examples, i.e. (1) spin precession; and (2) neutrino oscillation are discussed in detail in S&N 2.1, and will be a part of HW3. In particular, a physical observable you will encounter a lot in the future is the *transition probability*:

$$P(i \to f) \equiv |\langle f|e^{-\mathrm{i}\hat{H}t/\hbar}|i\rangle|^2 \tag{1.82}$$

which measures the probability that an initial state $|i\rangle$ evolves into a final state $|f\rangle$ after time t. This quantity is directly related to the outgoing beam intensity in scattering experiments.

The transition probability is the modulus square of the *propagator*

$$G(i, f : t) \equiv \langle f|e^{-\mathrm{i}\hat{H}t/\hbar}|i\rangle \tag{1.83}$$

which plays very important conceptual and computational roles in quantum mechanics. We will learn more about the propagator in HW3.

1.5.1 Schrodinger vs. Heisenberg pictures

Regarding the time dependence of physical observables, such as the expectation value of a Hermitian operator $\hat{O}$ at time t, there are two equivalent ways to approach this problem.

In the Schrodinger picture, where the state vector $|\psi\rangle$ evolves with time while the operator

$\hat{O}$ remains a constant:

$$|\psi_S(t)\rangle = \hat{U}(t,0)|\psi_0\rangle, \quad \hat{O}_S(t) = \hat{O} \tag{1.84}$$

In the Heisenberg picture, on the other hand, the state vector remains a constant while the operator evolves with time:

$$|\psi_H(t)\rangle = |\psi_0\rangle, \quad \hat{O}_H(t) = \hat{U}^\dagger(t,0)\hat{O}\hat{U}(t,0) \tag{1.85}$$

It is straightforward to show that both pictures yield the same physical observable

$$\langle O(t)\rangle = \langle\psi_S(t)|\hat{O}_S(t)|\psi_S(t)\rangle = \langle\psi_H(t)|\hat{O}_H(t)|\psi_H(t)\rangle = \langle\psi_0|\hat{U}^\dagger(t,0)\hat{O}\hat{U}(t,0)|\psi_0\rangle \tag{1.86}$$

In general, it is more convenient to use the Heisenberg picture to compute expectation values of an observable, while the Schrodinger picture is more convenient to extract time dependence of the state vector and its related properties, such as entanglement entropy.

1.5.2 Quantum Zeno effect

Consider a quantum system under repeated measurements of a physical observable $\hat{O}$ with the following spectral decomposition:

$$\hat{O} = \sum_q q\ |q\rangle\langle q| \tag{1.87}$$

Without loss of generality, let's assume the initial state of the system is $\hat{O}$ eigenstate $|0\rangle$ with eigenvalue q_0.

Now consider a generic Hamiltonian

$$\hat{H} = H_{0,0}|0\rangle\langle 0| + \sum_{q\neq 0}(H_{q,0}|q\rangle\langle 0| + h.c.) + \sum_{p,q\neq 0} H_{p,q}|p\rangle\langle q| \tag{1.88}$$

that governs the time evolution of the system. We used $h.c.$ as a shorthand notation for "hermitian conjugate".

TODO

1.6 Composite quantum systems (9/11)

- Refer to N&C 2.1.7, 2.2.8, 4.3; Tong 3.6. For generalized measurements, refer to N&C 2.2.6 and Tong 16.4.2

1.6.1 Tensor product

Previously we have focused on a single closed quantum system. When two closed quantum systems are combined together e.g. through some interactions b/w the two systems, how to describe such as composite quantum system? The last postulate addresses this issue, under the mathematical frameworks of *tensor product*:

Postulate 1.11. *The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and the system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.*

Specifically, consider two quantum systems with Hilbert space V of dimension m and W of dimension n, their composite system is described by the tensor product of the two Hilbert spaces:

$$V \otimes W \equiv \{|v\rangle \otimes |w\rangle \,|\, |v\rangle \in V, |w\rangle \in W\} \tag{1.89}$$

which is a Hilbert space of dimension mn. Linear operators on $V \otimes W$ are given by $A \otimes B$, where A is a linear operator on V and B is a linear operator on W, satisfying

$$(A \otimes B)|v\rangle \otimes |w\rangle = A|v\rangle \otimes B|w\rangle \tag{1.90}$$

An inner product on $V \otimes W$ is naturally defined as

$$(\sum_i a_i|v_i\rangle \otimes |w_i\rangle, \sum_j b_j|v'_j\rangle \otimes |w'_j\rangle) = \sum_{i,j} a_i^* b_j \langle v_i|v'_j\rangle\langle w_i|w'_j\rangle \tag{1.91}$$

One can therefore define a complete set of orthonormal basis $\{|v_i\rangle \otimes |w_j\rangle\}$ for $V \otimes W$, and the matrix representation of linear operators in $V \otimes W$ is given by

$$(A \otimes B)_{i,j} = A_{i_1,j_1} B_{i_2,j_2}, \quad A \in L_V,\; B \in L_W. \tag{1.92}$$

Since $A \otimes B$ is a matrix of dimension mn, where m and n are dimensions of matrices A and B respectively, the row and column index of matrix $A \otimes B$ can be written as

$$i = (i_1 - 1)n + i_2, \quad j = (j_1 - 1)n + j_2. \tag{1.93}$$

More properties of tensor product can be found in N&C section 2.1.7.

A *product state* in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$ is defined as

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle \tag{1.94}$$

It has NO quantum entanglement between subsystems 1 and 2.

With the help of tensor product, we can define Hamiltonian and unitary operators for a composite quantum system. We will study the transverse field Ising model for a 2-qubit system

$$H_{\text{TFIM}} = -JX_1 \otimes X_2 - h(Z_1 \otimes \hat{1}_2 + \hat{1}_2 \otimes Z_2) \tag{1.95}$$

as a demonstration. Due to a conserved quantity $Z_1 \otimes Z_2$ that commutes with the Hamiltonian, it can be diagonalized in the $Z_1 \otimes Z_2 = \pm 1$ subspaces respectively.

Name	Controlled-Z (CPHASE)	CNOT	SWAP
Unitary matrix	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Table 1.2: Common 2-qubit gates and associated unitary matrices.

Name	Toffoli (Controlled-CNOT)	Fredkin (Controlled-SWAP)
Unitary matrix	$\frac{1+Z_1}{2} \otimes \mathrm{CNOT}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$	$\frac{1+Z_1}{2} \otimes \mathrm{SWAP}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$

Table 1.3: Common 3-qubit gates and associated unitary matrices.

A good set of examples of unitary operators in a composite system are 2-qubit and 3-qubit quantum gates, shown in Table 1.2 and 1.3 respectively, such as a controlled-U gate in the 2-qubit system.

1.6.2 Generalized measurements

Composite systems play an important role in understanding general measurements in quantum mechanics, where we consider the measured system (Hilbert space $\mathbb{V}_s$) and an ancilla (Hilbert space $\mathbb{V}_a$) together as a composite quantum system, whose Hilbert space is the tensor product $\mathbb{V}_s \otimes \mathbb{V}_a$.

Consider an initial state

$$|\Phi\rangle \equiv |\psi\rangle_s \otimes |\phi\rangle_a \tag{1.96}$$

unitarily evolving under a Hamiltonian $\hat{H}$ of the composite system for time t. Next we perform a projective measurement on the ancilla, associated with observable $\hat{1}_s \otimes \hat{O}_a$, where Hermitian operator $\hat{O}$ has the following spectral decomposition:

$$\hat{O} = \sum_q q \ |q\rangle\langle q| \tag{1.97}$$

The basis vectors $\{|q\rangle\}$ forms a complete set of orthonormal basis for ancilla Hilbert space $\mathbb{V}_a$.

After this measurement, if the measurement outcome is q, the system collapses into a new state $M_q|\psi\rangle_s \otimes |q\rangle_a$ (unnormalized), where linear operator M_q is defined as

$$M_q \equiv \langle q|_a e^{-\mathrm{i}\hat{H}t/\hbar}|\phi\rangle_a \in L_{\mathbb{V}_s} \tag{1.98}$$

satisfying the completeness relation

$$\sum_q M_q^\dagger M_q = \hat{1}_s \tag{1.99}$$

The probability for measurement outcome q is given by

$$\Pr(q) = \mathrm{Tr}(M_q|\psi\rangle_s\langle\psi|_s M_q^\dagger) = \langle\psi|M_q^\dagger M_q|\psi\rangle \tag{1.100}$$

Clearly, the completeness relation (1.99) for operators $\{M_q\}$ guarantees the normalization of probability distribution:

$$\sum_q \Pr(q) = 1. \tag{1.101}$$

If we disregard the ancilla and examine this measurement from the viewpoint of the system, for measurement outcome q, initial state $|\psi\rangle$ becomes $M_q|\psi\rangle$, with a probability $\Pr(q)$ given in (1.100). This is known as a generalized measurement.

If the Kraus operator M_q is very close to identity matrix $\hat{1}_s$

$$M_q = \sqrt{1-\lambda_q}\hat{1} + \hat{O}_q, \quad \lambda_q \ll 1 \tag{1.102}$$

such a generalized measurement is known as a weak measurement because the measurement only modifies the initial state weakly in the sense that

$$\Big||\psi\rangle - M_q|\psi\rangle\Big| \ll 1 \tag{1.103}$$

This is the case when initial state $|\phi\rangle = |q\rangle$ and the evolution time t is much smaller than any energy scale in Hamiltonian $\hat{H}$.

2 Density matrix formulation of quantum mechanics

2.1 Density matrix and mixed states (9/16)

- Refer to N&C 2.4, 2.5; Tong 16.3.3, 16.4

If we think of a physical system (with Hilbert space A) and its environment (with Hilbert space B) combined together as a (composite) closed quantum system, any state therein is described by a state vector

$$|\psi\rangle = \sum_{i,j} \psi_{i,j} |a_i\rangle \otimes |b_j\rangle \tag{2.1}$$

where $\{|a_i\rangle | 1 \leq i \leq m = \dim A\}$ is a complete set of orthonormal basis for A, while $\{|b_j\rangle | 1 \leq j \leq n = \dim B\}$ is a complete set of orthonormal basis for B.

To study physical properties of system A, the composite state $|\psi\rangle$ in (2.1) is an overkill because it also contains all information about the environment, in which we are not interested at all. It turns out the following quantity, known as the *density matrix* (or *density operator*) for subsystem A

$$\hat{\rho} = \mathrm{Tr}_B\, |\psi\rangle\langle\psi| = \sum_{i_1,i_2=1}^{m} \rho_{i_1,i_2} |a_{i_1}\rangle\langle a_{i_2}|, \quad \rho_{i_1,i_2} = \sum_{j=1}^{n} \psi_{i_1,j}\psi^*_{i_2,j} \tag{2.2}$$

encodes all information about physical system A in composite state $|\psi\rangle$. As shown above it is obtained by tracing out subsystem B in state $|\psi\rangle$. Any physical observable of the system A corresponds to a Hermtian operator $\hat{O}_A \times \hat{1}_B$ in the composite system $A \times B$, and its expectation value is nothing but

$$\langle \hat{O} \rangle_A \equiv \langle \psi | \hat{O}_A \otimes \hat{1}_B | \psi \rangle = \mathrm{Tr}(\hat{\rho}\hat{O}) \tag{2.3}$$

Instead of directly computing the partial trace, there is another efficient way to obtain the density matrix (2.2), known as *Schmidt decomposition*, which we describe below. This approach is especially informative for understanding entanglement between subsystem A and B.

We can treat the wavefunction $\psi_{i,j}$ in (2.1) as a matrix of dimension $m \times n$, which has the following singular value decomposition (SVD):

$$\psi = U^\dagger D V, \quad U^\dagger U = V^\dagger V = \hat{1}, \quad D_{i,j} = \delta_{i,j}\lambda_i, \ \lambda_i \geq 0. \tag{2.4}$$

where U, V are two unitary matrices of dimensions $m \times m$ and $n \times n$ respectively. D is a $m \times n$ matrix whose matrix elements all vanish except for positive diagonal elements $\{\lambda_i > 0 | 1 \leq i \leq r,\ r \leq \min(m,n)\}$, known as singular values of matrix ψ. The positive integer $r \leq \min(m,n)$, is the rank of matrix $\hat{\psi}$.

Since unitary matrices implement the change of basis, the SVD procedure introduced a new set of orthonormal basis (known as the Schmidt basis) for both the system A and environment B:

$$|A_i\rangle = \sum_{i'} U^*_{i,i'} |a_{i'}\rangle, \quad |B_j\rangle = \sum_{j'} V_{j,j'} |b_{j'}\rangle \tag{2.5}$$

such that the state vector (2.1) can be written as

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |A_i\rangle \otimes |B_i\rangle, \quad \lambda_i > 0 \tag{2.6}$$

This is known as the Schmidt decomposition of state vector $|\psi\rangle$ in a composite Hilbert space, which is the tensor product of A and B. The rank r of matrix ψ is known as the Schmidt rank (or Schmidt number), and the positive numbers $\{\lambda_i > 0 | 1 \leq i \leq r\}$ are known as Schmidt coefficients.

Note that normalization of state vector $|\psi\rangle$ implies that $\{p_i \equiv \lambda_i^2\}$ leads to a normalized probability distribution:

$$\langle\psi|\psi\rangle = \sum_{i=1}^{r} \lambda_i^2 = 1 = \sum_{i=1}^{r} p_i \tag{2.7}$$

In the Schmidt basis $\{|A_i\rangle\}$, the density matrix (2.2) has the following diagonal form:

$$\hat{\rho} = \sum_{i=1}^{r} \lambda_i^2 |A_i\rangle\langle A_i| = \sum_{i=1}^{r} p_i |A_i\rangle\langle A_i| \tag{2.8}$$

which has the physical meaning as an ensemble $\{p_i, |A_i\rangle\}$ of pure states. p_i is the probability for the system A to fall into pure state $|A_i\rangle$.

Mathematically, a density matrix is defined as a linear operator $\hat{\rho}$ satisfying 2 conditions:

(1) **Trace condition**

$$\mathrm{Tr}\,\hat{\rho} = 1 \tag{2.9}$$

(2) **Positivity condition**

$$\hat{\rho} \geq 0 \tag{2.10}$$

The positivity of $\hat{\rho}$ already implies that it is a Hermitian operator i.e. $\hat{\rho} = \hat{\rho}^\dagger$.

All density matrices form a convex set in the sense that $x\rho_1 + (1-x)\rho_2,\ \forall\ 0 \leq x \leq 1$ is also a density matrix if $\rho_{1,2}$ are density matrices.

In the diagonal basis (2.8) of density matrix $\hat{\rho}$, if the Schmidt rank $r > 1$ i.e. there are at least two independent pure states in the ensemble, we call $\hat{\rho}$ a *mixed state*. If $r = 1$, instead we call $\hat{\rho} = |A_1\rangle\langle A_1|$ to be a *pure state*.

Theorem 2.1. *A density matrix $\hat{\rho}$ corresponds to a pure state if and only if*

$$\hat{\rho}^2 = \hat{\rho} \tag{2.11}$$

As a simplest example of Schmidt decomposition, we consider a single-qubit physical system A and a single-qubit environment B, where a pure state of the combined closed system has the following general form:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \tag{2.12}$$

where we have used $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$ (known as the computational basis) to label eigenstates of the Pauli-$\hat{Z}$ operator. We also used a shorthand notation for tensor product states:

$$|ab\rangle \equiv |a\rangle_A \otimes |b\rangle_B, \quad a, b = 0, 1. \tag{2.13}$$

In this case we have the matrix $\psi_{a,b}$ as

$$\hat{\psi} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{2.14}$$

for the SVD procedure.

Similar to the case of pure states, in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$, one can also define a *product state* for generic mixed states:

$$\rho_{\text{product}} = \rho_1 \otimes \rho_2 \tag{2.15}$$

and a *separate state*:

$$\rho_{\text{separable}} = \sum_i p_i \rho_{1,i} \otimes \rho_{2,i}, \quad \sum_i p_i = 1. \tag{2.16}$$

A product state features vanishing mutual information (see next section) between subsystems 1 and 2, while a separable state features vanishing quantum entanglement (e.g. in terms of entanglement negativity, see HW4) between sysbsystems 1 and 2.

2.2 Entanglement and quantum teleportation (9/18)

- Refer to N&C 1.3.6-1.3.7, 2.4.3; Tong 16.2.3, 16.3.2, 16.3.4. Use N&C 11.3 for further readings on entropy.

2.2.1 Entanglement entropy

In the Schmidt decomposition described in section 2.1, if the density matrix $\hat{\rho}$ is obtained by a partial trace over B from state ρ_{AB} in the composite system $A \otimes B$:

$$\hat{\rho}_A = \mathrm{Tr}_B\, \hat{\rho}_{AB} \tag{2.17}$$

it is called a *reduced density matrix*. Importantly, the reduced density matrix characterizes the correlation and entanglement between subsystem A and B, as we illustrate below.

Consider the following Hamiltonian in a composite system of two qubits A and B:

$$H = -JX_A \otimes X_B - hZ_A \otimes Z_B, \quad h, J > 0 \tag{2.18}$$

It can be simultaneously diagonalized with the conserved quantities $Z_A \otimes Z_B$ and $X_A \otimes X_B$. Its ground state with the lowest energy $E_0 = -h - J$ is given by

$$|Bell_0\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{2.19}$$

while the 3 excited states are given by

$$|Bell_1\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \tag{2.20}$$

$$|Bell_2\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \tag{2.21}$$

$$|Bell_3\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}. \tag{2.22}$$

These 4 states are known as *Bell states* or *EPR pairs*. They are all examples of bipartite *maximally entangled states* in the two-qubit system, in the sense that their reduced density matrix is proportional to the identity operator:

$$\hat{\rho}_A = \mathrm{Tr}_B\, |Bell_j\rangle\langle Bell_j| = \frac{1}{2}\hat{1}_A, \quad j = 0, 1, 2, 3. \tag{2.23}$$

Generally, a bipartite pure state $|\psi_{AB}\rangle$ is maximally entangled if and only if

$$\hat{\rho}_A \equiv \mathrm{Tr}_B\, |\psi_{AB}\rangle\langle\psi_{AB}| = \frac{1}{\mathrm{dim}A}\hat{1}_A \tag{2.24}$$

In order to quantitatively measure the quantum entanglement between two subsystems, we need to introduce the concept of von Neumann entropy for a density matrix $\hat{\rho}$:

$$S(\hat{\rho}) \equiv -\mathrm{Tr}\, \hat{\rho} \log \hat{\rho} \tag{2.25}$$

In the diagonal form (2.8) of density matrix $\hat{\rho}$, the von Neumann entropy defined above reduces to the Shannon entropy

$$S = -\sum_j p_j \log p_j \tag{2.26}$$

for a probability distribution $\{p_j\}$, an important concept in information theory. If we take the equilibrium density matrix $\hat{\rho} \propto e^{-\hat{H}/k_B T}$, the von Neumann entropy is nothing but the thermodynamic entropy of the corresponding quantum system described by Hamiltonian $\hat{H}$.

Now for a pure state $|\psi_{AB}\rangle$ in our bipartite system $A \otimes B$, if we consider the reduced density matrix $\hat{\rho}_A$ for subsystem A, its associated von Neumann entropy is called the *entanglement entropy*, because it quantitatively captures the entanglement (or more precisely, correlation) between subsystems A and B. The entanglement entropy has the following properties:

(1) Positivity: $S(\hat{\rho}) \geq 0$. In particular, $S(\hat{\rho}) = 0$ if and only if $\hat{\rho}$ is a pure state.

(2) Upper bound: $S(\hat{\rho}_A) \leq \log(\dim A)$, which is saturated by $\hat{\rho}_A = \hat{1}_A/\dim A$. This is why (2.24) is called a maximally entangled state.

(3) $S(\hat{\rho}_A) = S(\hat{\rho}_B)$, which comes directly from Schmidt decomposition of pure state $|\psi_{AB}\rangle$. If $S(\hat{\rho}_A) = S(\hat{\rho}_B) > 0$, we say that A and B subsystems are (bipartite) entangled in a pure state $|\psi_{AB}\rangle$.

(4) Subadditivity: $S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$ for a mixed state ρ_{AB}. The equality holds if and only if $\rho_{AB} = \rho_A \otimes \rho_B$, which is called a bipartite *product state*. This property allows for defining the *mutual information* between subsystems A and B

$$I(A:B) \equiv S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0 \tag{2.27}$$

which is a measure of entanglement (or quantum correlations) between A and B in a mixed state. The mutual information reduces to twice the entanglement entropy for a pure state.

(5) Araki-Lieb inequality: $S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$.

(6) Concavity: $S(\sum_i p_i \rho_i) \geq \sum_i p_i S(\rho_i)$ for any probability distribution $\{p_i\}$ and any density matrices $\{\rho_i\}$.

(7) Strong subadditivity: refer to N&C 11.4.

2.2.2 Quantum teleportation

The quantum entanglement between subsystems is an important resource to transfer quantum information. One striking example of such is the quantum teleportation. Consider a general pure state of a single qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{2.28}$$

It encodes an infinite number of classical bits because α, β are both generic complex numbers. In order for Alice to send the information encoded in this qubit to Bob, an infinite amount of classical bits will be needed if the information is sent via classical communication. Remarkably, the quantum entanglement allows Alice to teleport the qubit to Bob by only sending 2 classical bits using classical communication. The procedure is as follows. Alice has two qubits 1 and 2 where qubit 1 encodes the desired state $|\psi\rangle_1$, while Bob has a single qubit 3. Initially qubit 2 and qubit 3 are in a maximally entangled state (i.e. a Bell state), say $|Bell_0\rangle_{23}$. Alice first implements a CNOT gate on qubit 1 and 2 (qubit 1 being the controlled qubit), followed by a Hadamard gate on qubit 1:

$$|\psi\rangle_1 \otimes |Bell_0\rangle_{23} \xrightarrow{CNOT_{12}} \frac{(\alpha|00\rangle + \beta|11\rangle)|0\rangle + (\alpha|01\rangle + \beta|10\rangle)|1\rangle}{\sqrt{2}} \tag{2.29}$$

$$\xrightarrow{H_1} \frac{1}{2}\big[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)\big] \tag{2.30}$$

$$= \frac{1}{2}\big[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)\big] \tag{2.31}$$

Next, Alice makes measurements of Pauli $\hat{Z}$ operator on qubits 1 and 2, with registered measurement outcome $M_{1,2} = 0, 1$ (i.e. $\hat{Z}_{1,2} = (-1)^{M_{1,2}} = \pm 1$). After the measurement, Bob's qubit 3 collapsed into the following state:

$$|\psi_{M_1,M_2}\rangle = \alpha|M_2\rangle + \beta(-1)^{M_1}|1 - M_2\rangle \tag{2.32}$$

She further sends the two bits of measurment outcomes $M_{1,2}$ to Bob via classical communication. According to the two classical bits, Bob further implements a unitary operation $\hat{Z}^{M_1}\hat{X}^{M_2}$ on the 3rd qubit, and he will successfully recover the desired single qubit state (2.28) in qubit 3:

$$\hat{Z}^{M_1}\hat{X}^{M_2}|\psi_{M_1,M_2}\rangle = |\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{2.33}$$

2.3 Generalized measurements and decoherence

- Refer to N&C 2.2.6, 2.4.2, 8.2; Tong 16.4; Anastopoulos 8.2

2.3.1 Generalized measurements on a mixed state

Consider the product state $\rho_s \otimes \rho_a$ as the initial state of a composite system $\mathbb{V}_s \otimes \mathbb{V}_a$, consisting of a physical system s unentangled with the experimental apparatus a. We assume the apparatus density matrix ρ_a has the following spectral decomposition:

$$\rho_a = \sum_j p_j |j\rangle_a \langle j|_a, \quad \sum_j p_j = 1, \ 0 < p_j \leq 1. \tag{2.34}$$

where $\{|j\rangle_e\}$ is a complete set of orthonormal basis for the apparatus Hilbert space $\mathbb{V}_e$.

The system and apparatus evolve together under a unitary operator $\hat{U}$, before a projective measurement is performed for an observable (i.e. Hermitian operator) on the apparatus Hilbert space

$$\hat{O} = \hat{1}_s \otimes (\sum_q q \ |q\rangle_a \langle q|_a) \tag{2.35}$$

It's straightforward to show that for measurement outcome q, the initial state ρ_s of the physical system collapses into the following (unnormalized) state

$$\rho_q \propto M_{q,j} \rho_s M_{q,j}^\dagger \tag{2.36}$$

with a probability distribution

$$\Pr(q) = \sum_j \mathrm{Tr}(M_{q,j} \rho_s M_{q,j}^\dagger) \tag{2.37}$$

where the measurement operators $\{M_{q,j}\}$ are defined as

$$M_{q,j} \equiv \langle q|\hat{U}|j\rangle \sqrt{p_j} \tag{2.38}$$

satisfying the completeness relation

$$\sum_{q,j} M_{q,j}^\dagger M_{q,j} = \hat{1}_s \tag{2.39}$$

The above discussion is completely in parallel to section 1.6.2, merely generalizing everything to the context of mixed states. Therefore we can formulate the measurement postulate of quantum mechanics for density matrices:

Postulate 2b: A *generalized quantum measurement* is described by a collection $\{M_m\}$ of measurement operators satisfying the completeness relation:

$$\sum_m M_m^\dagger M_m = \hat{1}. \tag{2.40}$$

If the state of the quantum system is ρ immediately before the measurements, then the probability that result m occurs is given by

$$p(m) = \mathrm{Tr}(\rho M_m^\dagger M_m) \tag{2.41}$$

and the (unnormalized) state of the system after the measurement is $\rho_m \propto M_m \rho M_m^\dagger$.

2.3.2 Examples of generalized measurements

2.3.3 Measurements induce decoherence

In its eigenbasis $\{|j\rangle | 1 \leq j \leq \dim\mathbb{V}\}$, a mixed state (2.8) is an ensemble of pure states $\{|j\rangle\}$

$$\hat{\rho} = \sum_j p_j |j\rangle\langle j|, \quad \sum_j p_j = 1. \tag{2.42}$$

while a generic pure state in the same physical system can be written as a (normalized) superposition of the basis vectors $\{|j\rangle\}$:

$$|\psi\rangle = \sum_j \psi_j |j\rangle, \quad \sum_j |\psi_j|^2 = 1. \tag{2.43}$$

But there is a big difference between the ("incoherent") ensemble summation in the mixed state (2.42) and the ("coherent") superposition in the pure state (2.43), in the sense that the pure state contains more information in the relative phases of coefficients ψ_j that can be measured experimentally. To make it clear, we use the normalization condition in (2.43) to define the probability distribution $p_j \equiv |\psi_j|^2$, and the normalized pure state can be rewritten as

$$|\psi\rangle = \sum_j \sqrt{p_j} e^{\mathrm{i}\phi_j} |j\rangle, \quad \sum_j p_j = 1. \tag{2.44}$$

While the global phase factor is a redundancy with no physical significance, there are $(\dim\mathbb{V}-1)$ relative phases that can be detected by physical observables.

To illustrate, consider a single qubit example, where we use $(\hat{X}, \hat{Y}, \hat{Z})$ to label the 3 Pauli matrices. We consider a mixed state $\hat{\rho}$ and a pure state ψ, both having 1/2 probability for either $Z = +1$ or $Z = -1$. In the computational basis $|0\rangle \equiv |\uparrow\rangle$, $|1\rangle \equiv |\downarrow\rangle$ that diagonalizes $\hat{Z}$, the mixed state is written as

$$\hat{\rho} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{2.45}$$

and the pure state as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{\mathrm{i}\phi}|1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ e^{\mathrm{i}\phi} \end{pmatrix} \tag{2.46}$$

If we now measure another observable $\hat{X}$ for both states, we end up with different expectation values:

$$\mathrm{Tr}(\hat{\rho}\hat{X}) = 0, \quad \langle\psi|\hat{X}|\psi\rangle = \cos\phi. \tag{2.47}$$

exactly rooted in the relative phases and associated quantum inteferences in the pure state.

As a result, the reduction from a pure state (2.44) to a mixed state (2.42) is a process of decoherence, i.e. a reduction of quantum interference. This can be achieved by a projective measurement (see postulate 1.9), where the measurement outcome is unknown. Consider the implementation of a measurement for physical observable diagonal in basis $\{|j\rangle\}$

$$\hat{O} \equiv \sum_j O_j |j\rangle\langle j| \tag{2.48}$$

on the pure state (2.44), If we don't know the outcome, after a projective measurement, we should view the resulting state as a summation of projector $|j\rangle\langle j|$ weighted by probability

$p_j = |\langle j|\psi\rangle|^2$, and hence we end up with the mixed state (2.42). This means a quantum measurement will generally decohere a quantum state, even if we don't know the measurement outcome. This is in sharp contrast to a classical system.

A Homework 1 (due 9/4)

A.1 Franck-Hertz experiment

In the Franck-Hertz experiment (see Fig. 1.2 in lecture notes), the sharp drop of electric current through the vacuum tube at a voltage of 4.9 V corresponds to the absorption of electron energy by the mercury atoms in the vacuum tube, where the atomic energy level of mercury atoms go through a transition. Compute the wavelength of light associated with this atomic energy level transition, show that it is approximately 254 nm, as observed in optical emission of the same vacuum tube.

A.2 Davisson-Germer experiment

(1) *Bragg's law* Consider the diffraction of a wave by a lattice of atoms with a lattice constant of d between layers. Derive the Bragg's law

$$n\lambda = 2d\cos\frac{\theta}{2}, \quad n = 0, 1, 2, 3, \cdots \tag{A.1}$$

for the diffraction pattern.

(2) Consider the Davisson-Germer experiment illustrated in Fig. 1.5 of overleaf lecture notes. For $\theta = 50^o$, $d = 0.091$ nm for nickel crystals, and an electron beam accelerated by a $V = 54$ volts voltage, show that it satisfies the Bragg's law with $n = 1$.

In part (1) and (2), we have dealt with a simplified version of the actual Davisson-Germer experiment[3], assuming the electron beam was scattered by a three-dimensional simple cubic lattice. Below we will seriously consider the actual situation of Davisson-Germer experiment in history, where an electron beam was scattered by the (111) surface of a nickel crystal in which Ni atoms form a face-centered cubic (FCC) lattice. In this case, we need to first derive the general Bragg condition for two-dimensional diffraction peaks and apply it to the setup of Davisson-Germer experiment. Below we carry out this calculation step by step.

(3) For electrons of energy 54 eV in the Davisson-Germer experiment, their penetration depth into the nickel crystal is less than 1 nanometer, comparable to the FCC lattice constant $a = 3.52$ Å. This means as a zeroth order approximation, we can model the Davisson-Germer experiment as electrons scattered by the top layer of Ni atoms parallel to the crystal surface. In Davisson-Germer experiment the incident beam of electrons is perpendicular to the cleaved (111) surface of the nickel crystal. Can you draw the two-dimensional lattice formed on the (111) surface of a nickel crystal? Given the FCC lattice constant $a = 3.52$ Åin nickel cystals, what is the distance $\tilde{a}$ between two nearest Ni atoms on its (111) surface?

(4) For the two-dimensional (2d) plane parallel to the (111) surface, we can choose the following two orthonormal vectors as two basis vectors for the 2d plane:

$$\hat{e}_1 = (-1, 1, 0)/\sqrt{2}, \quad \hat{e}_2 = (-1, -1, 2)/\sqrt{6}, \quad \hat{e}_1 \times \hat{e}_2 = (1, 1, 1)/\sqrt{3}. \tag{A.2}$$

In this two-dimensional lattice on (111) surface, the periodic pattern of Ni atoms are described

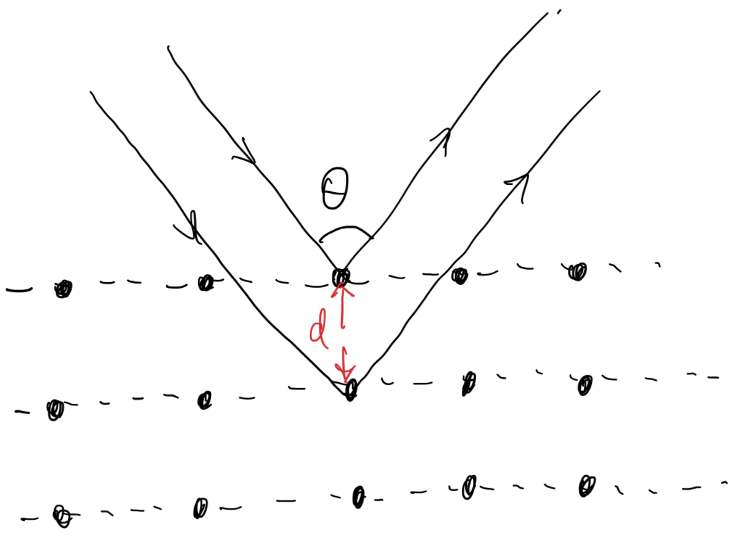

Figure A.1: Scattering of a wave by a lattice of atoms with a lattice constant d.

by two primitive vectors

$$\vec{a}_1 = \tilde{a}\hat{e}_1, \quad \vec{a}_2 = \tilde{a}\frac{-\hat{e}_1 + \sqrt{3}\hat{e}_2}{2}. \tag{A.3}$$

where $\tilde{a}$ is the distance $\tilde{a}$ between two nearest Ni atoms on its (111) surface. They are called primitive vectors in the sense that any lattice site position $\vec{R}$ in the 2d lattice is a summation of integer multiples of $\vec{a}_i$

$$\vec{R} = \sum_{j=1}^{2} m_j \vec{a}_j, \quad m_j \in \mathbb{Z}. \tag{A.4}$$

Most generally, the Bragg condition for diffraction peaks is given by

$$\sum_{\vec{R}} e^{\mathrm{i}\vec{q}\cdot\vec{R}} \neq 0 \Longleftrightarrow \vec{q}\cdot\vec{a}_j = 0 \mod 2\pi, \quad \forall\, j = 1, 2. \tag{A.5}$$

where $\vec{q}$ is the in-plane component of electron momentum transfer before and after scattered by the nickel crystal:

$$\vec{q} \equiv (\vec{k}_{\text{out}} - \vec{k}_{\text{in}}) - \frac{(1,1,1)}{3}(\vec{k}_{\text{out}} - \vec{k}_{\text{in}}) \cdot (1,1,1). \tag{A.6}$$

In later lectures we will come back to (A.5) when discussing Fourier transform as a change of basis (i.e. a unitary transformation). From (A.5), show that

$$\vec{q} = \sum_{j=1}^{2} n_j \vec{b}_j, \quad n_j \in \mathbb{Z} \tag{A.7}$$

and where the *reciprocal lattice primitive vectors* $\vec{b}_{1,2}$ are defined by the following conditions

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{i,j} \tag{A.8}$$

Note that $\vec{b}_{1,2}$ are both perpendicular to (1,1,1) as well. Express $\vec{b}_{1,2}$ using $\tilde{a}$ and $\hat{e}_{1,2}$.

(5) In the Davisson-Germer experiment, the incident momentum $\vec{k}_{\text{in}}$ is perpendicular to the (111) surface i.e. $\vec{k}_i = \frac{2\pi}{\lambda}(1,1,1)/\sqrt{3}$ where λ is the wavelength of electrons with a kinetic energy of 54 eV. For elastic scatterings responsible for the diffraction pattern, we have

$$|\vec{k}_i| = |\vec{k}_f| = \frac{2\pi}{\lambda} \tag{A.9}$$

In the experimental setup of Fig. 1.5 (left), using results (A.6)-(A.7) of the previous part, derive the following Bragg's law for Davisson-Germer experiment:

$$\lambda\sqrt{n_1^2 + n_1 n_2 + n_2^2} = \sqrt{\frac{3}{8}} a \sin\theta, \quad n_{1,2} \in \mathbb{Z} \tag{A.10}$$

Comment: specifically, if we choose $(n_1, n_2) = n(1,0)$ or $n(0,1)$ or $n(1,-1)$, Eq. (A.10) reduces to the more familiar one

$$n\lambda = d_{(111)} \sin\theta, \quad n \in \mathbb{Z}, \quad d_{(111)} = a\sqrt{3/8} \approx 2.16 \text{ Å}. \tag{A.11}$$

(6) Note that the in-plane momentum transfer $\vec{q}$ can be written as

$$\vec{q} = |q|(\cos\phi\hat{e}_1 + \sin\phi\hat{e}_2) \tag{A.12}$$

where ϕ is the azimuth angle in Fig. 1.5 of the lecture notes. Use relations (A.6)-(A.9) to explain the ϕ-dependent peaks in the right panel of Fig. 1.5.

(7) **Bonus** Can you qualitatively explain in Fig. 1.5 right panel, why are the peak heights at $\phi = \frac{\pi}{2} \mod 2\pi/3$ are different from those at $\phi = \frac{\pi}{6} \mod 2\pi/3$?

A.3 Hilbert space $\mathbb{L}_{\mathbb{V}}$ of linear operators on $\mathbb{V} = \mathbb{C}^n$

(1) Consider linear operators on complex vector space $\mathbb{V} = \mathbb{C}^n$, i.e. $n \times n$ complex matrices. Prove that they form a vector space, which we label as $\mathbb{L}_{\mathbb{V}}$. For the definition of a vector space, refer to Chapter 1.1 of Shankar.

(2) Prove that the following binary operation $(\cdot,\cdot)$ on $\mathbb{L}_{\mathbb{V}} \times \mathbb{L}_{\mathbb{V}}$ defined by

$$(A, B) \equiv \text{tr}(A^\dagger B), \quad A, B \in \mathbb{L}_{\mathbb{V}} \tag{A.13}$$

is an inner product. For the definition of an inner product, refer to Shankar 1.2 or N&C 2.1.4. This is known as the Hilbert-Schmidt inner product on operators.

(3) Using the above Hilbert-Schmidt inner product, show that 3 Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ defined below

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{A.14}$$

and the 2×2 identity matrix $\hat{1}$ together can form a complete orthonormal basis for the vector space $\mathbb{L}_{\mathbb{C}^2}$ of all 2×2 complex matrices.

(4) **Bonus** Prove the following *completeness relation of Pauli matrices*:

$$\sum_{i=x,y,z} (\sigma_i)_{a,b}(\sigma_i)_{c,d} = \vec{\sigma}_{a,b} \cdot \vec{\sigma}_{c,d} = 2\delta_{a,d}\delta_{b,c} - \delta_{a,b}\delta_{c,d} \tag{A.15}$$

Hint: use the conclusion of part (3), i.e. any 2×2 Hermitian matrix $\hat{M}$ can be written as

$$\hat{M} = \frac{\mathrm{Tr}(\hat{M})}{2}\hat{1}_{2\times 2} + \sum_{a=x,y,z} \frac{\mathrm{Tr}(\hat{M}\sigma_a)}{2}\sigma_a \tag{A.16}$$

A.4 Compton shift

Bonus Consider the Compton effect discussed in lecture notes (Fig. 1.1), where an X-ray photon was scattered by a stationary electron. Use energy and momentum conservations to derive the Compton shift formula:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta) \tag{A.17}$$

B Homework 2 (due 9/11)

B.0.1 Pauli matrices

(1) Diagonalize the 3 Pauli matrices and find out their eigenvalues and eigenkets:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{B.1}$$

(2) For an arbitrary function $f(\cdot)$ from complex numbers to complex numbers, if $\vec{n}$ is a normalized vector in three dimensions and θ is a real number, prove the following relation

$$f(\theta\vec{n}\cdot\vec{\sigma}) = \frac{f(\theta)+f(-\theta)}{2}\hat{1} + \frac{f(\theta)-f(-\theta)}{2}\vec{n}\cdot\vec{\sigma} \tag{B.2}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are three Pauli matrices.

Hint: a function $f(\hat{M})$ of a matrix (operator) $\hat{M}$ can be defined by the Taylor expansion:

$$f(\hat{M}) = \sum_{n=0}^{\infty} \frac{\hat{M}^n}{n!}\frac{\mathrm{d}^n f}{\mathrm{d}x^n}|_{x=0} \tag{B.3}$$

B.0.2 Linear algebra

(1) Use the Cauchy-Schwarz inequality to prove the triangle inequality

$$||u\rangle + |v\rangle| \leq ||u\rangle| + ||v\rangle|, \quad \forall\ |u\rangle, |v\rangle \in V \tag{B.4}$$

(2) A linear operator $\hat{P}$ is a projection operator (or a projector) if $\hat{P}^2 = \hat{P}$. Prove that the eigenvalues of a projector are all either 0 or 1.

(3) **Bonus** A projector has the following form:

$$\hat{P} = \alpha|1\rangle\langle 1| + \beta|2\rangle\langle 2| \tag{B.5}$$

where $|1\rangle$ and $|2\rangle$ are two normalized vectors in the Hilbert space that are linearly independent. What conditions should coefficients α and β satisfy in order for $\hat{P}$ defined above to be a projector (i.e. to satisfy $\hat{P}^2 = \hat{P}$)?

(4) Assume Hamiltonian H commutes with two linear operators, A_1 and A_2, which do not commute with each other:

$$[H, A_1] = [H, A_2] = 0, \quad [A_1, A_2] \neq 0 \tag{B.6}$$

Show that the eigenstates of H are in general degenerate.

(5) Assume two observables A_1 and A_2 anticommute with each other, i.e.

$$\{A_1, A_2\} = A_1A_2 + A_2A_1 = 0 \tag{B.7}$$

Show that for any eigenstate $|\psi_1\rangle$ of observable A_1 with a nonzero eigenvalue, its expectation value w.r.t. A_2 must vanish

$$\langle\psi_1|A_2|\psi_1\rangle = 0 \tag{B.8}$$

Comment: one such example is Pauli matrices $A_1 = \sigma_x$ and $A_2 = \sigma_y$.

(6) Consider two observables $A_{1,2}$ that anticommute with each other. Is it possible to have simultaneous eigenket of A_1 and A_2? Explain.

(7) **Bonus** For two linear operators satisfying $[\hat{A}, \hat{B}] = \alpha\hat{1}$ where $\hat{1}$ is the identity operator, prove the following identity:

$$[\hat{A}, e^{\hat{B}}] = \alpha e^{\hat{B}} \tag{B.9}$$

Hint: first show that

$$\hat{f}(x) = e^{-x\hat{B}}\hat{A}e^{x\hat{B}} = \hat{A} + x\alpha\hat{1} \tag{B.10}$$

(8) For a quantum-mechanical particle in one dimension, its position $\hat{x}$ and momentum $\hat{p}$ operators satisfy the canonical commutation relation:

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar = \mathrm{i}h/2\pi \tag{B.11}$$

Use the results of part (7) to show that for every $\hat{x}$ eigenstate $|x_0\rangle$ with eigenvalue x_0

$$\hat{x}|x_0\rangle = x_0|x_0\rangle \tag{B.12}$$

the following state

$$|x_0 + a\rangle \equiv e^{-\mathrm{i}\hat{p}a/\hbar}|x_0\rangle \tag{B.13}$$

is also an eigenstate of $\hat{x}$ with eigenvalue $x_0 + a$.

Comment: physically this means the linear operator $\hat{T}(a) \equiv e^{-\mathrm{i}pa/\hbar}$ translate the position by a, from x_0 to $x_0 + a$.

B.0.3 Consecutive measurements of incompatible operators

Two consecutive projective measurements, for Pauli matrices $\hat{\sigma}_x$ and $\hat{\sigma}_z$ respectively, are carried out one after another on a quantum state of a single qubit (i.e. a Hilbert space $\mathbb{V} = \mathbb{C}^2$). In the first procedure, we first measured $\hat{\sigma}_x$ with an outcome $+1$ and probability $p(X = +1)$, and then measured $\hat{\sigma}_z$ with an outcome $+1$ with probability $p(Z = +1|X = +1)$. In the second procedure, we switched the order of the two measurements, i.e. we first measured $\hat{\sigma}_z$ with an outcome $+1$ and probability $p(Z = +1)$, and then measured $\hat{\sigma}_x$ with an outcome $+1$ with probability $p(X = +1|Z = +1)$. It is found that the probabilites do no depend on the order of the two incompatible measurements, i.e.

$$p(X = +1) = p(X = +1|Z = +1), \quad p(Z = +1|X = +1) = p(Z = +1) \tag{B.14}$$

(1) Find a single-qubit pure state $|\psi\rangle$ that satisfies the condition (B.14).

(2) **Bonus** If the quantum state is a single-qubit mixed state, it is generally described by a density matrix of the following form:

$$\hat{\rho} = \frac{\hat{1} + \vec{r}\cdot\vec{\sigma}}{2}, \quad \vec{r} = (x, y, z) \in \mathbb{R}^3, \quad |\vec{r}| \leq 1. \tag{B.15}$$

Find those mixed states that satisfy the same condition (B.14).

Comment: you will need to learn about measurements on mixed states in N&C 2.4.1.

B.0.4 Quantum Zeno effect

Consider a single qubit evolving under the following Hamiltonian:

$$\hat{H} = -\mu_B B \hat{\sigma}_z \tag{B.16}$$

where $\mu_B = \hbar e/2m_e$ is the Bohr magneton. We define time T as follows:

$$T = \frac{\pi m_e}{eB} \tag{B.17}$$

Starting from the initial state

$$|\psi_0\rangle = |+\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}} \tag{B.18}$$

at time $t = 0$, we evolve the system by time T/n, followed by a measurement of $\hat{\sigma}_x$, and then repeat the same 2-step (evolution by $\frac{T}{n}$ + $\hat{\sigma}_x$ measurement) procedure n times. What is the probability at time T (after n such 2-step procedures) to produce the measurement outcome of $\sigma_x = +1$?

C Homework 3 (due 9/18)

C.0.1 Spin precession

(1) For a unit vector $\vec{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, diagonalize the matrix $\vec{n}\cdot\vec{\sigma} = \sum_{i=x,y,z} n_i\sigma_i$. Prove that it has two eigenvalues ± 1, and find out the two eigenvectors.

(2) We denote $|\vec{n}\rangle$ as the eigenvector of $\vec{n}\cdot\vec{\sigma}$ with eigenvalue $+1$. Starting from time zero, the state vector $|\psi(0)\rangle \equiv |\vec{n}\rangle$ evolves under the following Hamiltonian:

$$\hat{H} = -B\sigma_z \tag{C.1}$$

Prove that the following expectation value of $S_z = \hbar\sigma_z/2$

$$\langle S_z(t)\rangle \equiv \langle\psi(t)|S_z|\psi(t)\rangle \tag{C.2}$$

does not change with time.

(3) Compute the time-dependent expectation values of S_x and S_y, and show that they oscillate with time.

(4) Compute the transition probability that the state vector $|\vec{n}\rangle$ returns to itself after time t:

$$P(\vec{n} \to \vec{n} : t) \equiv |\langle\psi(t)|\psi(0)\rangle|^2 \tag{C.3}$$

C.0.2 Neutrino oscillation

Read the neutrino oscillation in S&N 2.1. Using the energy eigenvalues

$$E_i = \sqrt{p^2c^+m_i^2c^4} \approx pc(1 + \frac{m_i^2c^2}{2p^2}) \tag{C.4}$$

prove the probability

$$P(\nu_e \to \nu_e) = 1 - \sin^2(2\theta)\sin^2\big[\frac{(m_1^2 - m_2^2)c^4t}{4E\hbar}\big], \quad E \approx pc. \tag{C.5}$$

C.0.3 Propagator

(1) Consider the following propagator for Hamiltonian $\hat{H}$:

$$G(i, f; t) \equiv -\mathrm{i}\,\langle f|e^{-\mathrm{i}\,\hat{H}t/\hbar}|i\rangle \tag{C.6}$$

Show that its Fourier transform

$$G(i, f : \omega + \mathrm{i}0^+) \equiv \lim_{\epsilon\to 0^+}\int_0^\infty e^{\mathrm{i}\,(\omega+\mathrm{i}\,\epsilon)t}G(i, f : t) \tag{C.7}$$

is equal to

$$G(i, f : \omega + \mathrm{i}0^+) = \langle f|\frac{1}{\omega + \mathrm{i}0^+ - \hat{H}}|i\rangle \tag{C.8}$$

where we have set $\hbar = 1$.

(2) **Bonus** Now consider a quantum particle in one dimension, by choosing i, f to be position eigenstates $|i\rangle = |x\rangle$, $|f\rangle = |y\rangle$. Show that

$$G(x, y; t) = -\mathrm{i} \sum_n \psi_n^*(x)\psi_n(y)e^{-\mathrm{i} E_n t/\hbar} \tag{C.9}$$

where E_n and $\psi_n(x)$ are the eigenvalues and eigenstates of Hamiltonian $\hat{H}$.

(3) **Bonus** Show that the imaginary part of the frequency-domain propagator is given by

$$\Im G(x, x; \omega + \mathrm{i}0^+) = -\pi \sum_n |\psi_n(x)|^2 \delta(\omega - E_n/\hbar) \tag{C.10}$$

where $\delta(x)$ is a Dirac delta function.

C.0.4 2-qubit Hamiltonian

Consider a closed quantum system consisting of two spin-1/2 particles $\{\vec{S}_i = \frac{\hbar}{2}\vec{\sigma}_i | i = 1, 2\}$, i.e. two qubits. The Hamiltonian of the system is given by the *Heisenberg model*

$$\hat{H} = J\vec{S}_1 \cdot \vec{S}_2 = \sum_{a=x,y,z} JS_{1,a} \otimes S_{2,a} \tag{C.11}$$

(1) Using the matrix representation of Pauli matrices, show the commutation relation for each qubit:

$$[S_{j,a}, S_{j,b}] = \mathrm{i}\hbar\epsilon_{abc}S_{j,c}, \quad j = 1, 2; \quad a, b, c = x, y, z \tag{C.12}$$

where ϵ_{abc} is the Levi-Civita symbol.

(2) Prove that each component of the total spin

$$S_a^{\text{tot}} \equiv S_{1,a} \otimes \hat{1}_2 + \hat{1}_1 \otimes S_{2,a} \tag{C.13}$$

commutes with the Hamiltonian:

$$[S_a^{\text{tot}}, \hat{H}] = 0, \quad a = x, y, z. \tag{C.14}$$

The total spin of the system is defined as

$$S_{\text{tot}}^2 = \vec{S}^{\text{tot}} \cdot \vec{S}^{\text{tot}} = \sum_a S_a^{\text{tot}} S_a^{\text{tot}} \tag{C.15}$$

Show that it commutes with both the Hamiltonian $\hat{H}$, and S_a^{tot}, $a = x, y, z$.

(3) Show that if the Hamiltonian $\hat{H}$ commutes with another Hermitian operator $\hat{O}$ (known as a conserved quantity or a constant of motion), for any two eigenstates $|\psi_1\rangle, |\psi_2\rangle$ of operator $\hat{O}$ with different eigenvalues $\lambda_1 \neq \lambda_2$, the following matrix must vanish

$$\langle\psi_1|\hat{H}|\psi_2\rangle = 0 \tag{C.16}$$

In other words, Hamiltonian $\hat{H}$ is a block diagnoal matrix in the eigenbasis of the conserved quantity $\hat{O}$. This is known as the selection rule in quantum mechanics.

(4) Diagonalize the Hamiltonian $\hat{H}$ and find its energy levels. Find the simultaneous eigenstates of $\hat{H}$, S_z^{tot} and S_{tot}^2, in the orthonormal basis of $\{|\sigma_{1,z} = \pm 1\rangle \otimes |\sigma_{2,z} = \pm 1\rangle\}$. Note: these 4 eigenstates can be properly chosen to form the *Bell basis* of 2-qubit Hilbert space $\mathbb{C}^4$. (You are allowed to use computer programs, but please show your codes if you do so.)

(5) **Bonus** Can you explain the reason for the degenerate and non-degenerate energy levels, using what you have learnt from Problem 2.(4) in Homework 2?

D Homework 4 (due 9/25)

D.0.1 Bloch ball for single-qubit mixed states

(1) Show that an arbitrary density matrix for a single-qubit mixed state can be written as

$$\hat{\rho} = \frac{\hat{1} + \vec{r} \cdot \hat{\vec{\sigma}}}{2} \tag{D.1}$$

where $\vec{\sigma}$ are 3 Pauli matrices, and $\vec{r}$ is a 3-dimensional vector such that $|\vec{r}| \leq 1$. This vector $\vec{r}$ is known as the *Bloch vector* for the state $\hat{\rho}$.

Hint: a density matrix $\hat{\rho}$ is defined to satisfy two properties: $\hat{\rho}$ is positive (i.e. it can be diagnonalized and all eigenvalues are non-negative) and $\mathrm{Tr}\rho = 1$.

(2) Show that a state ρ is pure if and only if $|\vec{r}| = 1$ i.e. $\vec{r}$ is a unit vector.

Hint: a density matrix $\hat{\rho}$ is a pure state if and only if $\hat{\rho}^2 = \hat{\rho}$.

(3) Consider the eigenstate $|\vec{r}\rangle$ of operator $\vec{r} \cdot \vec{\sigma}$ with eigenvalue $+1$, where $\vec{r}$ is a unit vector. Show that in this case, its density matrix $\rho = |\vec{r}\rangle\langle\vec{r}|$ is exactly given by Eq. (D.1) with $\vec{r}$ being a unit vector. This is known as the Bloch sphere.

(4) Compute the von Neumann entropy $S(\rho) = -\operatorname{Tr} \rho \log \rho$ for the mixed state (D.1). Plot it as a function of $|\vec{r}|$.

D.0.2 Entanglement entropy

(1) If ρ_1 and ρ_2 are two density matrices, show that

$$\hat{\rho}(x) \equiv x\hat{\rho}_1 + (1-x)\hat{\rho}_2, \quad 0 \leq x \leq 1. \tag{D.2}$$

is also a density matrix.

(2) **Bonus** Assume that $\hat{\rho}(x)$ in (D.2) is always invertible for any $0 \leq x \leq 1$, i.e. its inverse matrix $[\hat{\rho}(x)]^{-1}$ always exists. Prove that its von Neumann entropy is a concave function of x, i.e.

$$\frac{\mathrm{d}^2 S(\hat{\rho}(x))}{\mathrm{d}x^2} \leq 0 \tag{D.3}$$

(3) **Bonus** Use the concave property (D.3) to prove the concavity of von Neumann entropy

$$S(\sum_j p_j \hat{\rho}_j) \geq \sum_j p_j S(\hat{\rho}_j) \tag{D.4}$$

where $\{p_j\}$ is a probability distribution and $\{\hat{\rho}_j\}$ is a set of density matrices.

Hint: for a concave function $f(x)$ satisfying $f'' \leq 0$, we always have $f(px_1 + (1-p)x_2) \geq pf(x_1) + (1-p)f(x_2)$ for $0 \leq p \leq 1$.

(4) Consider a pure state $|\psi\rangle_{AB}$ in a composite quantum system with Hilbert space $A \otimes B$. Show that for the reduced density matrices

$$\rho_A = \mathrm{Tr}_B\, |\psi_{AB}\rangle\langle\psi_{AB}|, \quad \rho_B = \mathrm{Tr}_A\, |\psi_{AB}\rangle\langle\psi_{AB}| \tag{D.5}$$

they share the same von Neumann entropy

$$S(\rho_A) = S(\rho_B) \tag{D.6}$$

Therefore either $S(\rho_A)$ or $S(\rho_B)$ can be defined as the entanglement entropy.

(5) Consider a composite system of two qubits A and B. Can you write down the generic form of a maximally entangled state $|\psi_{AB}\rangle$, satisfying that the reduced density matrix has a maximal entanglement entropy?

Hint: for a maximally entangled 2-qubit state, we must have

$$\hat{\rho}_A = \hat{\rho}_B = \frac{1}{2}\hat{1}. \tag{D.7}$$

so that $S(\rho_{A,B}) = \log 2$.

(6) For a pure state in a composite system of 2 qubits A and B:

$$|\psi_{AB}\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \tag{D.8}$$

show that this state is (bipartite) entangled if and only if

$$\alpha\delta - \beta\gamma \neq 0. \tag{D.9}$$

Hint: a pure state is (bipartite) entangled if and only if its bipartite entangled entropy is nonzero:

$$S(\rho_A) = S(\rho_B) > 0 \tag{D.10}$$

(7) **Bonus** This problem is about *tripartite entanglement*. Show that in a 3-qubit system, there are pure states that cannot be written in the following form:

$$|\psi\rangle = \sum_{j=1}^{2} \lambda_j |\alpha_j\rangle_1 \otimes |\beta_j\rangle_2 \otimes |\gamma_j\rangle_3. \tag{D.11}$$

where $\{\alpha_j\}$, $\{\beta_j\}$, $\{\gamma_j\}$ are any complete basis for $\mathbb{C}^2$. Find one such state.

D.0.3 Entanglement negativity

When we consider a mixed state ρ_{AB} defined in a composite system $A \otimes B$, the entanglement entropy $S(\rho_A)$ should be replaced by the mutual information

$$I(A:B) \equiv S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \tag{D.12}$$

Clearly it reduces to (twice) the entanglement entropy $2S(\rho_A)$ in the special case when ρ_{AB} is a pure state.

In a generic mixed state, the mutual information (D.12) is incapable of distinguishing classical correlation betweem A and B from the quantum entanglement. To capture bipartite quantum entanglement in a mixed state ρ_{AB}, we need to introduce new entanglement measures. One such quantity is (logarithmic) entanglement negativity:

$$E_{\mathcal{N}}(\rho_{AB}) \equiv \log ||\rho_{AB}^{T_A}||_1 \tag{D.13}$$

where T_A denotes a partial transpose on subsystem A.

The trace norm $||\cdot||_1$ for a density matrix ρ is defined as

$$||\rho||_1 \equiv \mathrm{Tr}\sqrt{\rho^\dagger \rho} = \sum_i |\lambda_i| \tag{D.14}$$

where $\{\lambda_i\}$ are the eigenvalues of density matrix ρ.

Obviously the logarithmic negativity $E_{\mathcal{N}}$ vanishes if all eigenvalues of $\rho_{AB}^{T_A}$ are non-negative.

(1) Show that

$$E_{\mathcal{N}}(\rho_{AB}) = \log(1 + 2\sum_{\lambda_i<0} |\lambda_i|) \tag{D.15}$$

where $\{\lambda_i\}$ are all eigenvalues of matrix $\rho_{AB}^{T_A}$.

(2) Now consider a composite system of two qubits, one qubit in A and one qubit in B. Show that for the Bell state $|Bell_0\rangle$ in this composite system, the logarithmic negativity is $\log 2$.

(3) **Bonus** Next consider a mixed state, which is the equilibrium density matrix at inverse temperature $\beta = J/k_B T$ (J is a constant of the dimension of energy)

$$\rho_{AB} \propto e^{-\beta Z_A \otimes Z_B} \tag{D.16}$$

where $\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the 3rd Pauli matrix. Show that its logarithmic negativity (D.13) vanishes, while the mutual information (D.12) is nonzero. Plot its mutual information as a function of $\beta > 0$.

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